Developer Guide

ScheduleFlow is a Java 25 console application for university students planning coursework around fixed commitments. For commands and user examples, see the User Guide.

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Acknowledgements

ScheduleFlow uses Gradle and the Shadow plugin to build and package its executable JAR, JUnit 5 for Java tests, and Checkstyle for automated style checks. Dependency versions are recorded in `build.gradle` and the Gradle wrapper configuration. These are development tools; ScheduleFlow does not call their websites while managing a student's data.

This guide adapts the organization and enhancement-explanation pattern of the AddressBook Level 3 Developer Guide. That credit concerns documentation structure; the console design described here uses ScheduleFlow's actual classes. The UML figures use PlantUML and the SE Education PlantUML tutorial. The repository's local skills record

the SE Education Java conventions, Git conventions, and CS2113 code-quality guidance used during development.

The code-authorship allocation identifies maintenance responsibilities. Reassigned whole-file tags do not establish who originally implemented an enhancement; Git history retains that provenance. Before release, contributors must supply credits for any additional copied or adapted material so its source and extent can be acknowledged accurately.

Setting up, getting started

Import the repository into IntelliJ as a Gradle project and select SDK 25. Keep Java source encoding at UTF-8. Use the committed Gradle wrapper; Node.js 22 or later is needed for process tests, not for running the application. See build and verification for commands and manual testing for isolated JAR launches.

Design

The following views separate responsibilities in the existing packages; the Implementation section traces the workflows that connect them.

The figures use the course's class and sequence notation. Solid open-headed arrows show retained references; dashed open-headed arrows show use or creation. A hollow triangle on a dashed line points to an implemented interface. Multiplicities and reference roles appear at the target end. Java records are marked «record» and their stored fields are private; omitted accessors remain available in the code. Sequence calls use filled heads and returns use dashed lines. A narrow activation represents one invocation, including time spent waiting for nested calls. Folded-corner notes explain deliberately omitted detail; dotted open-headed arrows point from each note to the element it describes. Colours identify responsibilities consistently across figures, independently of the team's ownership allocation.

System overview and collaboration boundaries

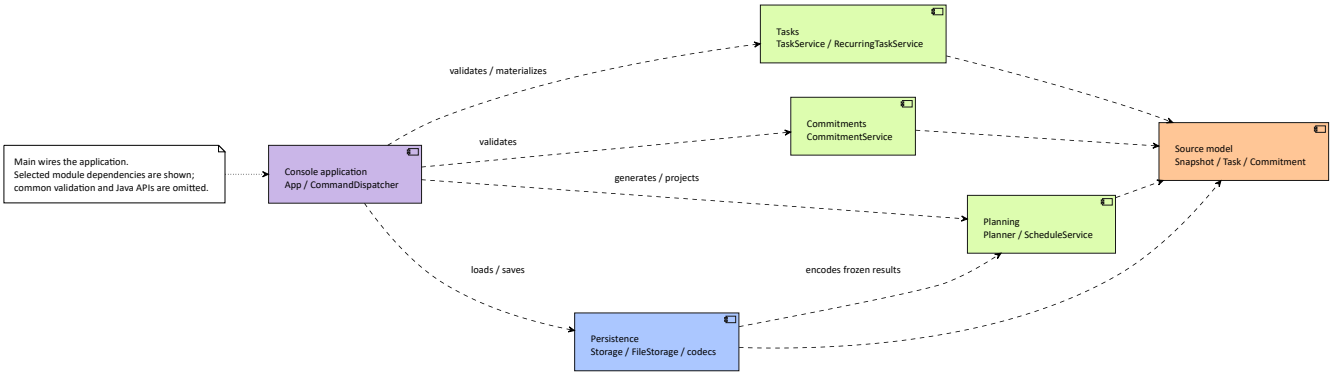


Figure 1. Component architecture and selected source dependencies.

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The console application coordinates the domain services and persistence. Services produce immutable candidates; the dispatcher saves and then publishes successful changes. Storage also depends on planning values to encode a frozen result. This overview omits shared validation and Java library dependencies. The class views below expand these components.

Area	Main entry points	Code owner
Application shell and presentation	`Main`, `App`, `AppState`, `ConsoleUi`, `TextRenderer`, `Command`, `CommandResponse`	Ka How (`ka-how`)
Tasks and all command parsing	`TaskService`, `TaskEdit`, `Task`, `CommandParser`, `ParseException`, shared validators	Nickhil (`nickpatt`)
Commitments and all command dispatch	`CommitmentService`, `Commitment`, `CommandDispatcher`	Darren (`Darren50`)
Planning, hours, policy and recurrence	`planning/`, `StudyHours`, `StudyWindow`, `RecurringTaskService`, `RecurringTask`, `TaskOccurrence`	Gabriel (`GabrielYuSP`)
Persistence and aggregate validation	`storage/`, `Snapshot`	Ernest (`ernestgoh-god`)

Code authorship

The whole-file `@@author` tags and the agreed version 2 allocation define ownership, including nested types and helpers. Cross-component calls do not split ownership. Tags record the allocation, not historical authorship. See source ownership, test ownership and the RepoSense tag reference.

Console application and presentation

The shell connects text interaction to the domain without putting console I/O inside services. The class view shows the shell's production types and the external collaborators needed to understand the command path. Associations denote retained references; dashed dependencies denote creation or use. `...` marks omitted parameters. Service, codec, clock and transport-use dependencies are omitted to keep this view focused.

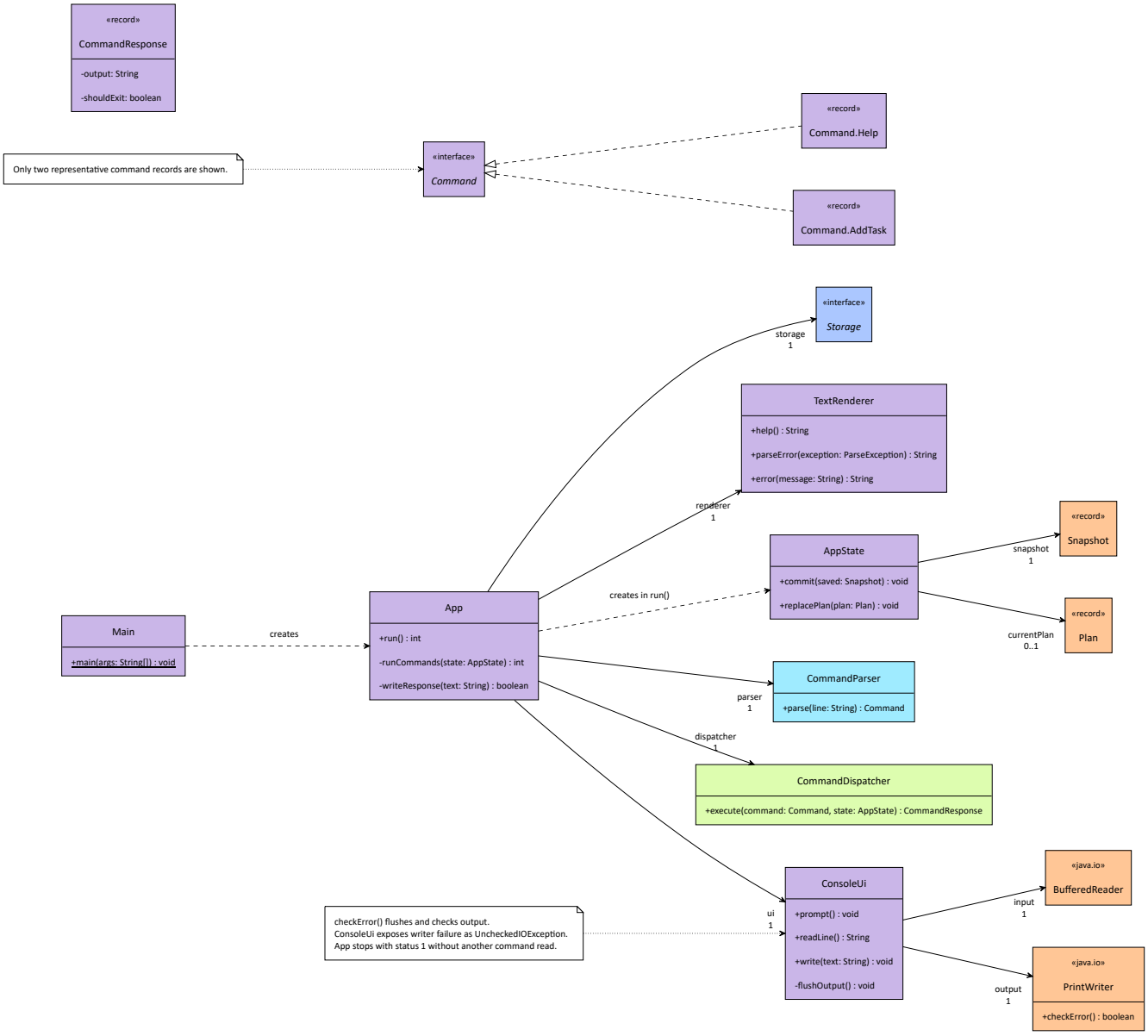


Figure 2. Console shell, command values and collaborators.

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`Main` supplies UTF-8 streams, `Clock.systemDefaultZone()` and `FileStorage` for `data/scheduleflow.txt` relative to the launch folder. It creates `App` and turns its returned status into process termination. `App` receives `Storage`, `Clock` and `ConsoleUi`, then constructs the parser, renderer and dispatcher with their services. Constructors do no I/O; loading begins in `run()`. `AppState` is local to that run, rather than an `App` field, so command execution receives the session's live state explicitly.

`ConsoleUi` owns line reads, prompt/response writes and flushing. Its retained reader/writer are shown in Figure 2. `flushOutput()` uses `PrintWriter.checkError()` to flush and detect suppressed write failures, exposing them as `UncheckedIOException`. `App` catches that exception only around console writes and returns status 1; unrelated component failures still propagate. No public signatures change. `TextRenderer` returns formatted text; it does not read the clock, perform console I/O or mutate state. `AppState` retains a `Snapshot` and an optional matching `Plan`. `commit(saved)` publishes a saved candidate and clears the plan. `replacePlan(plan)` accepts an equal source snapshot, rejecting a stale source before changing state, even with assertions disabled.

`Command` is a sealed interface whose nested records carry requests, for example `Help`, `AddTask` and `ScheduleToday`. These values have no execution method: syntax belongs to Nickhil's parser and domain rules to the services. `CommandResponse(output, shouldExit)` carries the dispatcher's formatted result and termination

decision back to `App`. The diagram shows only representative records; the two unavailable range transports execute atomic batch additions; active UI-022 through UI-027 cover their confirmations, rejection and restarts.

Task model and command parsing

Candidate scope: the Task sections describe the current source and its task/parser regression tests, not the published v1.0 JAR. Build this checkout when verifying the candidate; the release review records validation and remaining integration gates. The implementation explanation concerns the Task lifecycle enhancement; the later whole-file allocation also assigns the complete parser to Nickhil, but does not establish who originally wrote every parser or integration method.

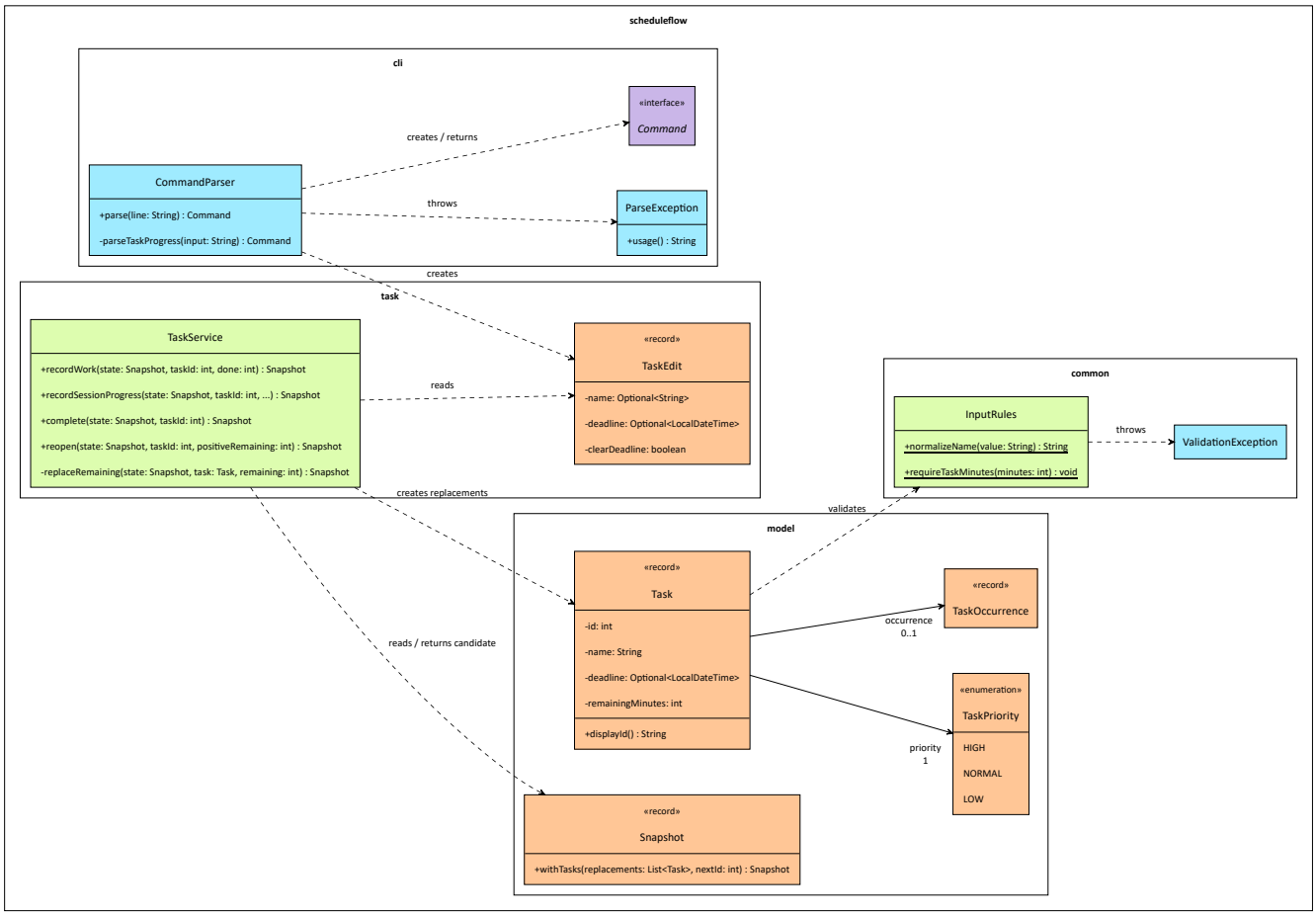


Figure 3. Task model, lifecycle service and parser dependencies.

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The two groups have different responsibilities. The stateless `CommandParser` matches the entire input and constructs Ka How's nested `Command` records. It does not look up IDs, read a clock, save data or invoke a service. `ParseException` preserves a message and the recognized command's usage hint. This applies to all command families, not just tasks. Both unavailable range transports have separate dispatcher/service acceptance; parser tests establish syntax only.

`TaskService` reads Ernest's immutable `Snapshot` and returns a validated candidate or the identical source for a no-op. It never owns live application state. `Task` is a record containing a stable ID, normalized name, optional deadline, exact remaining minutes, `TaskPriority`, and optional external `TaskOccurrence` provenance. Zero remaining means retained completion. `TaskEdit` distinguishes omitted replacements from an explicit API-level deadline clear. `InputRules` centralizes name and duration validation; `ValidationException` reports rejected domain values.

Dashed arrows denote creation/use dependencies, not lifetime ownership. `Task` retains value references to its priority and optional provenance. Gabriel owns template generation and the provenance type; Nickhil's service preserves that provenance when changing an already-generated task. Snapshot validation and storage remain external. The diagram intentionally omits repetitive accessors, parser patterns and constructor overloads.

Commitments and command execution

This component has two jobs. It models the student's fixed commitments, and it executes every parsed command as one transaction: validate, save, then publish. The class view shows commitment and dispatch types and the collaborators the dispatcher holds. Solid arrows are retained references; dashed arrows are use or creation. Colours are shared with the other views; contributor names remain in the ownership table, outside the UML model.

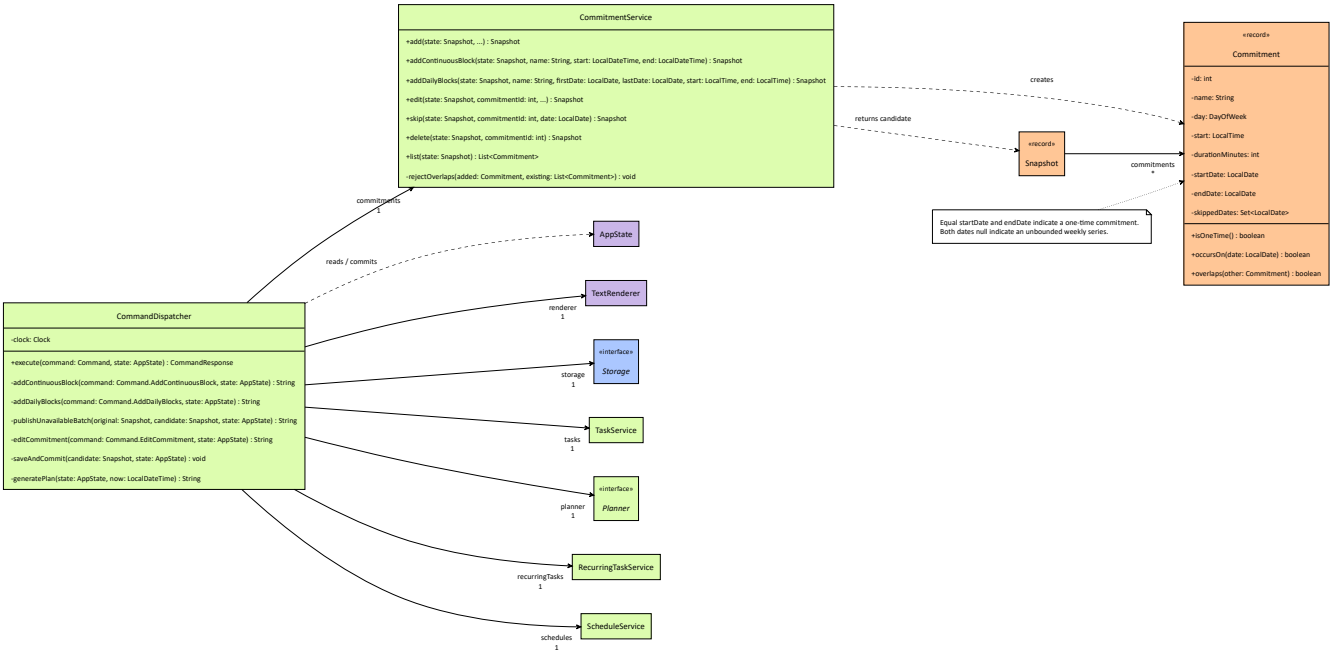


Figure 4. Commitment service and dispatcher collaborators.

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Type	Responsibility
`Commitment`	Immutable record of one fixed commitment. Its compact constructor validates the name, half-hour start, positive 30-minute-multiple duration, same-day end, date range and skipped dates. `occursOn(date)` and `overlaps(other)` are the only calendar-membership and conflict rules used elsewhere.
`CommitmentService`	Stateless operations (`add`, `addContinuousBlock`, `addDailyBlocks`, `edit`, `skip`, `delete`, `list`) that take a `Snapshot` and return a complete candidate `Snapshot` or an ordered view. It never saves, publishes or reads the clock.
`CommandDispatcher`	Routes each nested `Command` record to one handler, captures the command time once, and owns the save-before-publication transaction for every mutating command, not only commitments.

A `Commitment` takes one of three shapes, all stored in the same record:

Shape	`startDate` / `endDate`	Occurs on
Unbounded weekly series	both `null`	every day

Shape	`startDate` / `endDate`	Occurs on
Bounded weekly series	ordered dates containing day	every day within the inclusive range
One-time commitment	equal dates	that single date (its day is derived from it)

Weekly shapes may also hold `skippedDates`, which `occursOn` excludes. Two commitments overlap only when their times intersect, they share a weekday and at least one unskipped date falls in both. This lets separate semesters reuse a time slot while still rejecting real clashes.

The dispatcher separates three steps that other components keep apart too. A service validates and returns a candidate, but nothing changes yet. `Storage.save` then makes the candidate durable, using Ernest's atomic persistence. Only after that does `AppState.commit` publish it and clear the now-stale plan. The dispatcher reuses other members' services rather than duplicating their rules: task changes come from Nickhil's `TaskService`, recurring occurrences and plans from Gabriel's `RecurringTaskService` and `Planner`, and output text from Ka How's `TextRenderer`. Importing those types does not make them part of this component.

Planning, schedule projection and recurrence

`EarliestDeadlinePlanner` creates chronological half-hour slots from the captured command time through the rolling calendar-year horizon. It excludes recurring commitments, skips zero-remaining completed tasks, sorts dated tasks by deadline, priority then numeric ID, followed by undated tasks by priority/ID, and assigns each eligible task to its earliest free slots. Overdue and out-of-horizon tasks keep their full work estimate as unallocated; eligible tasks retain any sessions that fit before their deadline. Undated tasks use the same finite horizon without overdue/outside-horizon classification. Adjacent slots for one task merge within a day. Source task estimates remain unchanged.

`ScheduleService` projects the frozen plan across the complete configured daily window. Each slot is classified as busy, assigned to a task, reserved for a break, before the generation cutoff, or free, in that order. Merging preserves commitment and task identities. `Plan`, `StudySession`, `UnallocatedWork`, `ScheduleEntry` and `ScheduleView` validate their own fields and retain immutable result lists; allocation relationships remain the planner's responsibility. Fixed-date planner and projection tests verify deadline order, conservation of minutes, cutoff rounding, clipping, midnight commitments and allowed schedule dates.

`Planner.validate(Plan)` is the pure cross-record boundary. It checks the frozen source and generation time, not the current clock, without regenerating work. `EarliestDeadlinePlanner.generate` calls it before returning every result.

Ernest's codecs call this boundary before encoding and after decoding frozen plans; raw `Plan` construction retains its lightweight contract.

`Planner` is implemented by `EarliestDeadlinePlanner`. `TimeRules` defines cutoff and horizon calculations. `StudyHours` / `StudyWindow` supply daily availability and `StudyPolicy` supplies optional work/break limits. The dispatcher calls `RecurringTaskService` before planning; the allocator does not generate templates or occurrences itself.

Study hours and optional policy come from the plan's source snapshot. A date override takes precedence over weekday hours, which take precedence over the default. Recurrence is prepared outside the planner: the dispatcher asks `RecurringTaskService` to materialize a candidate before calling `Planner.generate`. The planner uses each occurrence's original release time and has no dependency on recurrence, storage or the live clock. A recurrence template owns its stable R ID, weekly deadline pattern, optional inclusive end date and processing cursor. Each generated task retains its template ID and original deadline, from which its one-week release is derived. Advancing the cursor past missed dates prevents backfilling and recreation after restart or deletion of a generated task.

The class view focuses on the planner and daily schedule projection boundaries. `Task` and `Commitment` are source records owned by their respective modules; `Snapshot` is the aggregate owned by the persistence component. Arrows show dependencies or contained results, not ownership of those external types.

The class view follows the UML Reference Sheet's operation/attribute visibility, dashed dependency and realization arrows, navigable associations, role names and multiplicities. Underlined members are static; `<<record>>` identifies Java record classes. The planner's optional policy association represents its constructor fallback, while the snapshot's policy is the saved setting. Shared immutable records use associations rather than exclusive-lifetime composition. Only selected members and the window-aware `eligibleStart` overload are shown.

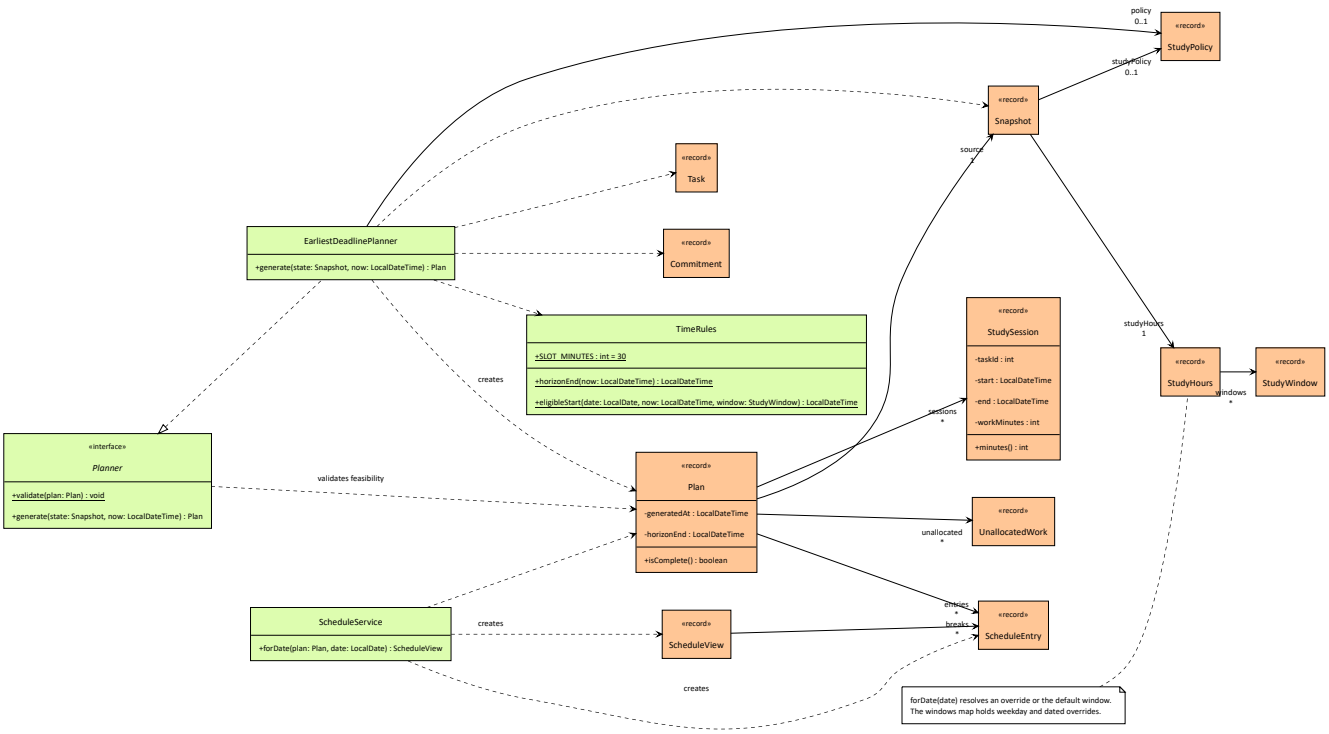


Figure 5. Planning and schedule projection structure.

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Generation first classifies unfinished work: overdue and beyond-horizon tasks retain their full remainder; other work competes for chronological slots. Dated tasks are ordered by deadline, priority and numeric ID, followed by undated tasks ordered by priority and ID. A slot must fit wholly inside that date's study window, start after the rounded generation cutoff and any occurrence release, end no later than the task deadline, and not overlap a fixed commitment. Required breaks reserve capacity too.

For example, at Monday 10:07 with an 08:00-22:00 window, a 10:00-10:30 class and a 90-minute task due at 11:30, the rounded cutoff is 10:30. The task gets the two remaining slots through its deadline: 60 minutes scheduled and 30 reported as insufficient capacity. Its source estimate remains 90 minutes. This earliest-deadline greedy policy is predictable and gives earlier deadlines first access to scarce time; it does not optimize for fewest sessions or most fully completed tasks.

The sequence view follows a successful `plan` command from time capture through plan publication. The planner obtains the horizon, resolves the daily cutoff for each date, and evaluates half-hour slots. `TimeRules` is a class participant because its methods are static; its internal horizon checks are omitted. The `Planner` interface is also shown as a `<<class>>` participant for its static `validate` call, following the reference sheet's class-level message

notation. Ordinary participants use `name:Class` labels. Solid calls, dashed returns, nested self-call activations and bracketed `loop` / `opt` guards show the successful path. Exception paths and local helper calls remain omitted. The dispatcher renders and saves the plan before publishing the candidate snapshot and current plan.

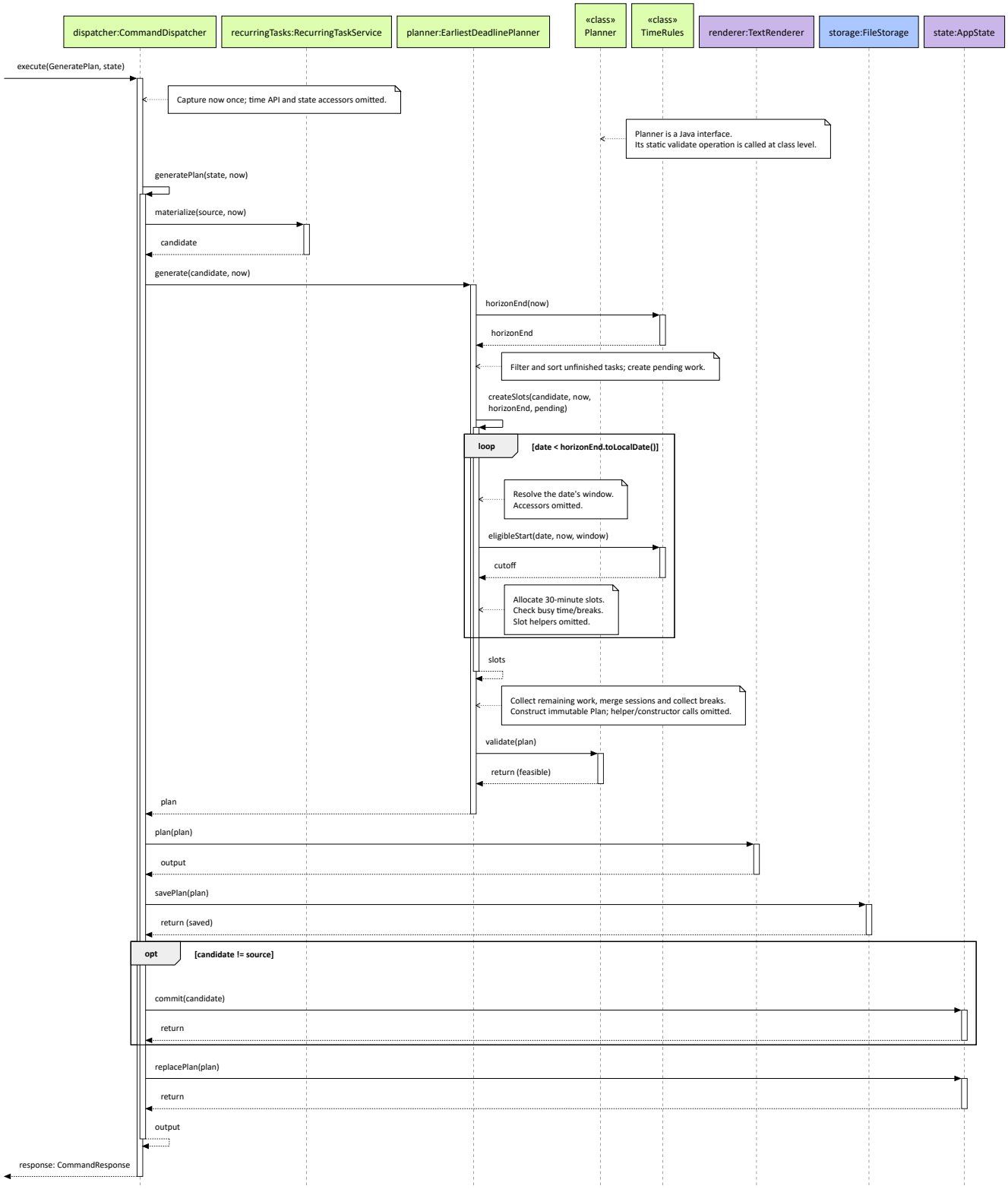


Figure 6. Successful plan generation, saving and publication.

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Persistence and the Snapshot aggregate

This section describes the current source implementation, not the older v1.0 JAR. The persistence boundary separates filesystem access, text conversion and aggregate validation:

Type	Responsibility
`Storage`	Defines complete snapshot and frozen-plan loading/saving contracts for the application.
`FileStorage`	Resolves one target path, reads UTF-8 and replaces that file atomically. Construction performs no I/O.
`FileStorage.FileOperations`	Package-private save-operation seam; normal instances delegate to `Files`, while storage tests inject faults without global filesystem changes.
`StateCodec`	Converts versioned source records without opening files.
`PlanCodec`	Converts a frozen result together with its complete source; it is package-private.
`Snapshot`	Copies lists and validates the complete persisted aggregate.
`StorageException`	Reports rejected loads/saves while retaining underlying causes where available.

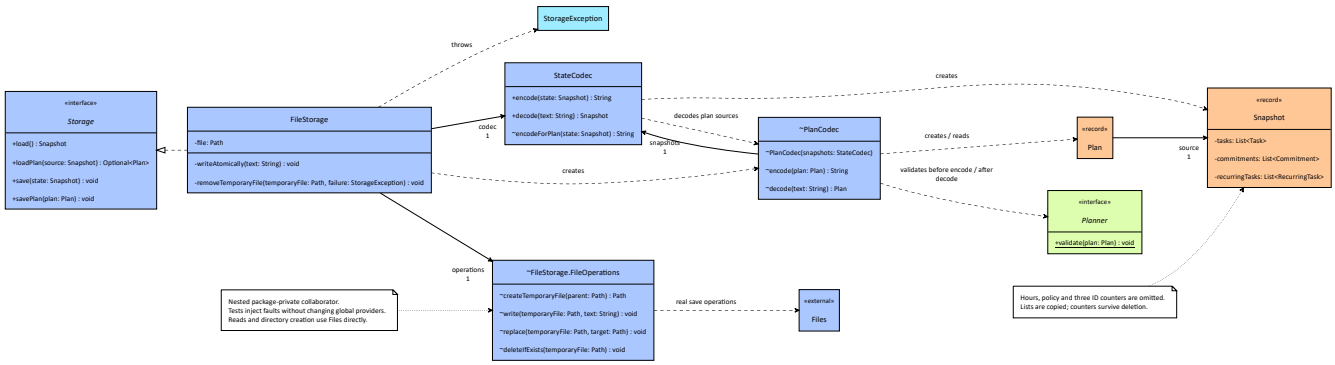


Figure 7. Persistence and the source aggregate.

PlantUML source · PNG. Dashed arrows denote use; solid arrows denote retained references. Other model dependencies appear as field types. `~PlanCodec` has package visibility; its members shown with `~` are also package-private.

`Snapshot` retains tasks, commitments, study hours, recurring templates, optional policy and three independent next-ID counters. Its validation rejects duplicate IDs, counters not above retained IDs, overlapping commitments and inconsistent occurrence provenance/cursors. Validated hours/policy values must be present in their required containers. Retained provenance may refer to a deleted template, provided its allocated ID remains below the next R counter. Lists retain their order and are immutable; deleting all records does not reset counters.

A `Snapshot` is source data. A `Plan` additionally freezes generation time, sessions, unallocated work and breaks against that source. Ka How's `AppState` holds the session's live snapshot and optional plan; storage does not publish that state or regenerate schedules.

`PlanCodec` calls Gabriel's pure `Planner.validate(Plan)` boundary before encoding and after decoding. Allocation rules remain in planning: infeasible intervals, overlaps, incorrect remainders and unbalanced work reject the complete plan with `StorageException` retaining the validation cause. Valid alternative frozen allocations remain unchanged; validation never reads today's clock.

Implementation

These sections describe the important state transitions and failure boundaries. Proposed console operations are explicitly labelled.

Each section traces one enhancement and links to the others instead of repeating them. A typical mutating command crosses all five:

1. Console interaction reads and parses the line and later prints the response.
2. Task lifecycle, commitment or planning services validate it and return a candidate.
3. The dispatcher transaction saves the candidate before publishing it.
4. Atomic persistence makes that save all-or-nothing.

Recoverable console interaction and deterministic rendering

A mistyped command should leave the student able to correct it in the same session. The shell implements this by distinguishing recoverable command failures from failures that prevent the session from continuing. Its starting points are `App.run()` and `runCommands()`.

Startup and the command loop

`run()` calls `Storage.load()`, creates `AppState`, and requests `Storage.loadPlan(loaded)`. A returned plan is checked through `replacePlan`. Neither restoration nor rendering regenerates the schedule. A `StorageException` during either load produces `TextRenderer.error(message)` and status 1 before any prompt or command read. Starting empty is allowed only for a genuinely missing store; corrupt data must not be silently replaced. File validation and atomic replacement are explained in persistence.

After successful startup, each iteration flushes the exact `scheduleflow> ` prompt, calls `ConsoleUi.readLine()`, and handles one complete line. Blank input immediately begins the next iteration, without parsing, dispatching or saving. For a nonblank line, the parser returns a `Command`; the dispatcher executes it against the same `AppState`. `ConsoleUi.write(response.output())` adds the final newline and flushes the response before `shouldExit()` is inspected.

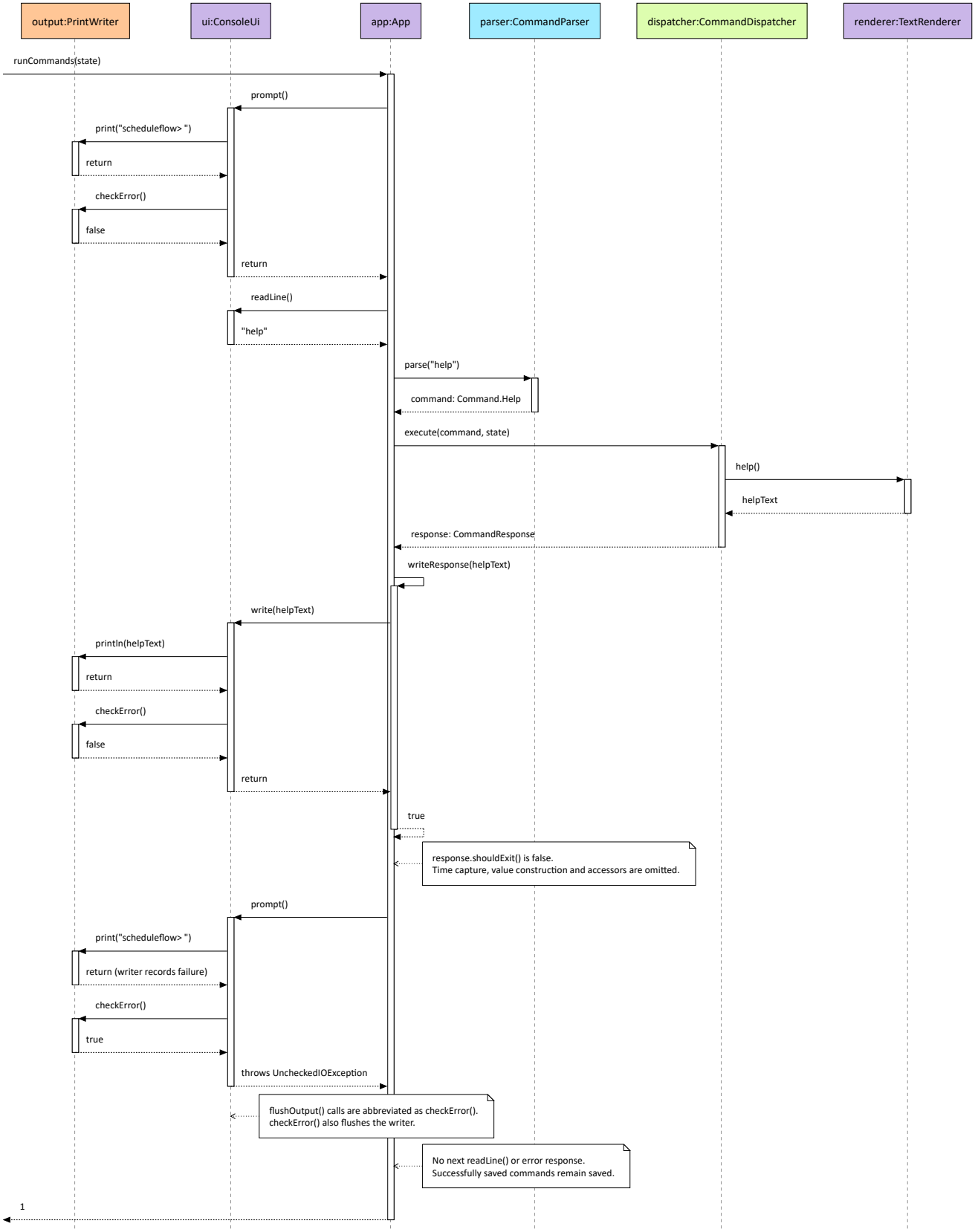


Figure 8. Help command followed by a failed prompt.

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Figure 8 follows one successful `help` command and then a failed prompt after restoration. The `runCommands` activation remains open while nested calls execute, and ends at its return of status 1. The failed prompt prevents the next input read. `writeResponse` catches output failure while writing a response; the prompt has its own narrow catch. The diagram abbreviates `ConsoleUi.flushOutput()` as its `checkError()` call, which flushes and checks the

writer's error flag. EOF bypasses parsing and dispatch. Blank input, syntax errors, validation/save failures and explicit `exit` are separate paths described below. Value construction and accessors are omitted. Mutation and publication order is shown in command transactions.

For example, after adding `commitment add n/Lecture day/MON start/1000 d/60` to an empty store, parsing supplies `Command.AddCommitment`. The dispatcher obtains a candidate, saves and publishes it, then calls `TextRenderer.commitmentAdded`. Only the returned `CommandResponse` reaches the console. Entering the same overlapping commitment again yields a validation error and another prompt. Correcting the interval to start at `1100` succeeds with C2; the rejected attempt consumed no ID. The shell owns the interaction, while the service and dispatcher establish candidate validation and rollback.

Failure boundaries and termination

Event	Shell behavior	Session outcome
`ParseException`, for example `help extra`	`parseError` prints `Error: MESSAGE` and `Expected: USAGE`; dispatch is not reached.	Next prompt.
`ValidationException`, for example an overlapping commitment	`error` prints one explanation; no success response is written.	Next prompt.
`StorageException` during command execution	`error` reports the rejected save; the dispatcher retains the previous live state and plan.	Next prompt; the user can retry.
`exit`	The dispatcher returns goodbye with `shouldExit = true`; `App` prints it once.	Status 0, no extra save or prompt.
EOF (`readLine()` returns `null`)	`App` writes `TextRenderer.exit()` without invoking parser or dispatcher.	Status 0, no save.
Input `IOException`	`run()` prints `Error: Cannot read input: MESSAGE`.	Status 1.
Output failure during a prompt, response, goodbye or error report	`ConsoleUi` detects the writer's error flag. `App` stops without reading another command or reporting the error to the failed writer.	Status 1; any already-saved command stays saved.

Unexpected programming exceptions propagate. In particular, a null dependency or a stale plan is not converted to an ordinary input error. Invalid range requests use domain validation; valid batches save once before publication. Extra process arguments are rejected by `Main` before loading, with `Error: Usage: java -jar scheduleflow.jar (no arguments)` and status 1. `App` returns status codes so tests and embedded callers need not terminate their JVM; only `Main` calls `System.exit`. If argument-error output itself fails, `Main` still exits with status 1 without loading data or attempting another write. Output-failure diagnostics follow the existing opt-in logging configuration; default launches keep stderr quiet.

Deterministic presentation

Renderer methods return LF-separated strings with no trailing newline. `ConsoleUi` supplies the final platform newline; tests normalize CRLF only at the process boundary. Explicit UTF-8 streams in `Main` preserve names such as `学习 Café` even when the JVM's default encoding differs. Date/time formatters use `uuuu-MM-dd`, `uuuu-MM-dd HH:mm` and `HH:mm` with `Locale.ROOT`; weekday labels explicitly use English. Validated commitment endpoints at midnight display as `24:00` instead of wrapping to `00:00`.

Formatting preserves the supplied task order and schedule entry boundaries. Weekly commitments are grouped Monday to Sunday, followed by one-time entries ordered by date; within each group the service's supplied ordering is retained. The renderer does not allocate, clip or merge sessions. `schedule(view)` uses the view's frozen

generation time and rows, adds a cutoff note if UNPLANNED appears, and reuses the whole-plan unallocated-work footer rather than filtering it to the selected date.

Exact work and reserved time have different meanings: a 45-minute task may occupy a 60-minute interval. Plan summaries sum `StudySession.workMinutes` and `UnallocatedWork.minutes` using `long` totals; TASK schedule rows show the wall-clock interval duration. Presentation must preserve that distinction, including when totals exceed a signed 32-bit integer.

Design considerations

The following is an analysis of the implemented design. Keeping rendering as returned text allows exact-output tests without redirecting global streams; injected readers/writers, storage and clocks allow controlled session tests. Printing directly inside each service would require fewer intermediate values, but would mix domain work with console framing and make isolated tests harder. The current design instead requires the dispatcher to choose the appropriate formatter and return `CommandResponse` consistently.

Catching the three expected command exceptions keeps correction local to one iteration. Catching every runtime exception would also keep the loop running, but could conceal a broken internal invariant. Propagating unexpected failures keeps them visible to developers. `PrintWriter` suppresses checked write errors, so `ConsoleUi` explicitly checks its flag after each prompt or response. Because that flag retains no original cause, the resulting `UncheckedIOException` contains a descriptive `IOException`, rather than a fabricated device cause. Output failure ends the session; attempting to print another error could fail again. It does not roll back a successful save: restart and inspect saved records before repeating a command whose confirmation could not be delivered.

`AppTest` covers startup, syntax/domain/save recovery and failed-save retries; `AppStateTest` covers source-plan consistency. `ConsoleUiTest` checks flushes and framing, `MainTest` checks process status/UTF-8/quiet defaults, and `TextRendererTest` checks exact text, midnight, large totals and locale changes. Fixed clocks, temporary files and isolated logger/locale changes keep these checks independent of the developer's saved data. See console verification and manual console tests.

Deterministic failing writers cover both write and flush failures. Application tests reject any input read after a failed prompt/response, check single-attempt startup/input/command-error reporting and goodbye failure, and reload a real saved mutation after its response fails. Child-JVM tests exercise the actual `Main` wiring with failed stdout, including argument rejection and quiet stderr. An unchecked I/O defect from storage still propagates instead of becoming a console status. These are controlled faults, not real broken-pipe or OS proof.

`MainTest` also checks corrupt saved-plan metadata and a directory at the saved-file path: both must exit with status 1 before prompting and preserve existing bytes. The directory-error assertion requires one complete error line and an absolute path that identifies the expected target using `Files.isSameFile`. File identity handles both Windows short/long names and macOS working-directory aliases without requiring the parent and child JVMs to spell the path identically. Separate child JVMs verify class-based logging configuration and explicitly set CLI/application logger levels remain effective without changing command output. The logging probes exist only on the test classpath, and assertions are enabled for these startup/error paths. Each child JVM uses an English locale for stable diagnostic severity text and keeps logger configuration inside that process.

Exact task work and retained task lifecycle

The enhancement lets a student record actual work without losing task identity or confusing remaining work with scheduled capacity. Suppose T1 has 90 minutes remaining. A reported 61-minute session at 25% credits **15**, leaving **75**. The percentage describes that session, not the original task estimate. Planning may reserve 90 minutes for the resulting 75-minute estimate, but does not alter it. See allocation for capacity accounting and the candidate scope for release status and contribution boundaries.

Parsing and domain validation

`task progress T1 session/61 percent/25` becomes `Command.RecordTaskSession(1, 61, 25)`. Its direct-work counterpart becomes `Command.RecordTaskWork`. Full-line matching rejects missing, reordered, repeated or extra prefixes. Positive decimal tokens reject signs, leading zeroes, non-ASCII digits and `int` overflow; progress explicitly permits zero credit and percentages 0..100. The parser rejects percentages above 100, while the service repeats its domain bounds for direct API callers. `ParseException` contains the relevant usage; a well-formed but unknown T ID instead produces `ValidationException` during execution.

Dates use strict `uuuu-MM-dd HHmm`, minute precision and years 0001..9999. `Locale.ROOT` avoids machine-locale grammar changes. Syntax validation never decides whether a deadline is future: the dispatcher supplies one captured `now` to operations needing time. Likewise `schedule today` remains unresolved until dispatch. This boundary makes the same Task service usable by tests and non-console callers without pretending a parsed command is already valid.

Exact progress and immutable candidates

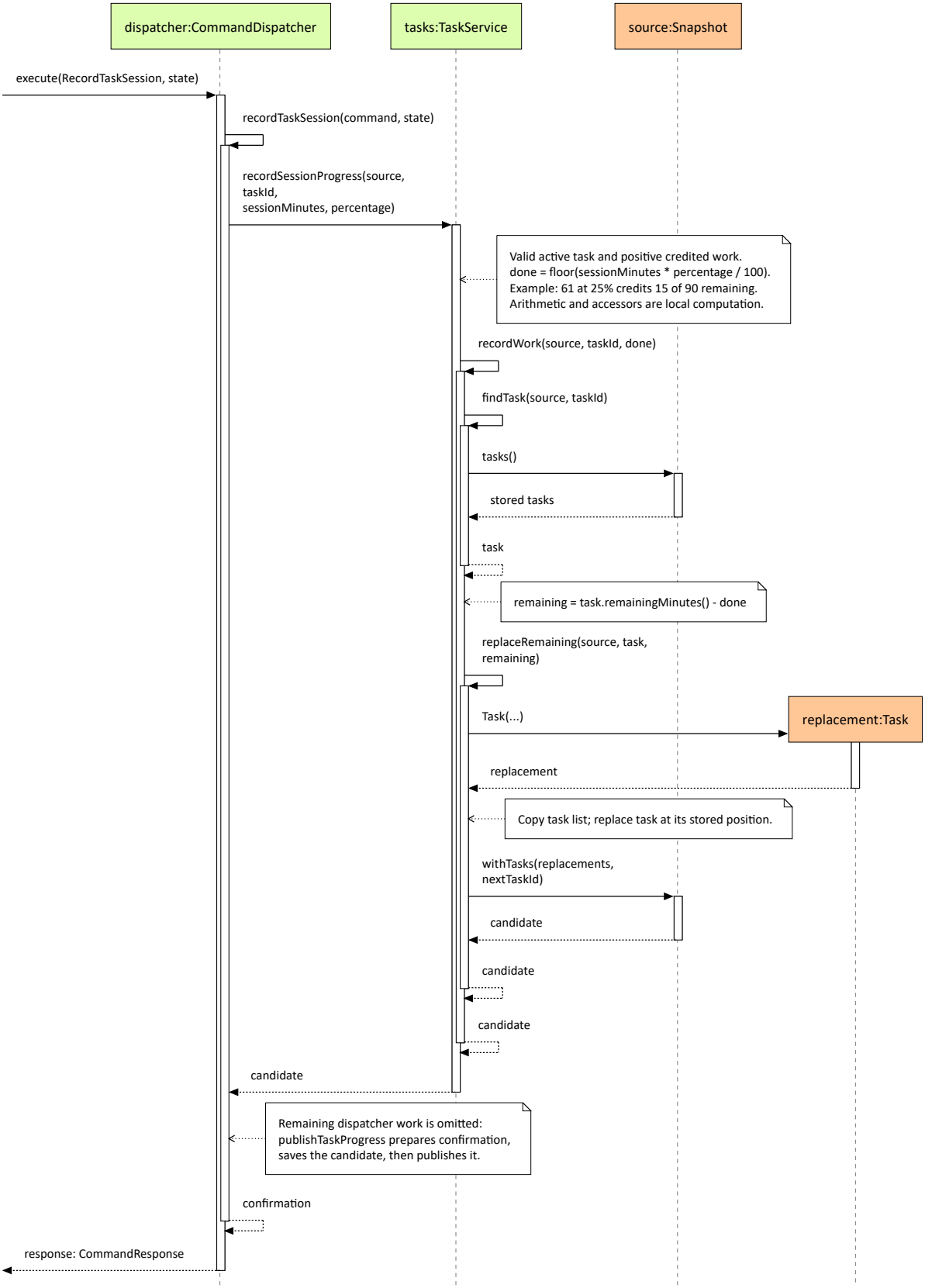


Figure 9. Positive session progress and task replacement.

PlantUML source · PNG

Figure 9 follows a valid 61-minute session at 25% on an active task with 90 minutes remaining. The 15-minute credit creates a new `Task`, whose header and lifeline begin at construction. `Task(...)` omits constructor parameters so the creation arrow reaches the object header; the copied fields are explained below. Dispatcher persistence and publication calls are summarized in a note. Zero-credit no-ops, invalid input and save failures are separate paths described below.

`recordSessionProgress` first validates positive session minutes and an integer percentage in 0..100. It computes `(int) ((long) sessionMinutes * percentage / 100)`. Integer division floors the nonnegative result; promotion before multiplication avoids overflow even for `Integer.MAX_VALUE` session minutes. The credit then passes through `recordWork`, the common authority for direct and session input. It resolves the existing task, rejects a completed record, and requires credit between zero and that task's remaining estimate. Overreported work is rejected, not clamped to zero remaining.

`replaceRemaining` returns the **identical Snapshot instance** when no minutes change. Otherwise it copies the task list, replaces the selected record at its original position, and calls `Snapshot.withTasks` with the unchanged next T ID. The new Task copies name, optional deadline, priority and occurrence provenance; the aggregate helper preserves other records, all counters, templates/cursors, hours and policy. The original remains valid and unchanged. TaskService neither saves nor publishes this candidate and does not invalidate or regenerate a plan.

The external dispatcher derives actual credited minutes from the before/after estimates and prepares Ka How's confirmation. For a changed candidate it invokes Darren's transaction; zero-credit identity no-ops skip saving and preserve the current plan. A save failure publishes nothing, so retrying applies the work once. Two *successful* submissions record work twice: there is no duplicate-command detection. Durable restart behavior belongs to Ernest's persistence.

Completion, reopening and corrections

Operation	Domain result and boundary
`complete`	Retains the same task with zero remaining. Repeated completion is an identity no-op.
`reopen`	Requires a completed task and an explicit positive estimate; no earlier estimate is silently restored. Overdue/undated records may reopen.
`reviseRemaining`	Replaces an active estimate, allowing increases; it does not credit completed work. An equal estimate is a no-op.
`edit`	Requires a supplied name/deadline change or explicit API clear. It preserves work/status; a normalized equal result is a no-op.
`delete`	Permanently removes a record, without resetting or reusing IDs. It is not completion.

The model permits zero work, but creation, revision and reopening require positive whole minutes. Completed tasks reject progress even for `done/0`. Their records remain in `task list`; the dispatcher implements the completed-only view using the service's numeric-ID list. A generated task has the same lifecycle, but its original release and template provenance survive every replacement. These mutations neither generate occurrences nor modify Gabriel's templates. Deleting a template also retains its generated tasks. Progress, completion, reopening and edits continue to work on those records without the live template; their original occurrence release remains unchanged even when the visible deadline changes. No deleted template is recreated by a task mutation.

Only a changed, present deadline is checked against `now < deadline <= TimeRules.horizonEnd(now)`. Renaming an overdue task or retaining its old deadline therefore works. Empty edits and simultaneous clear/replacement requests fail atomically. Optional deadlines and priority mutation exist at the API/storage layer; console addition still

requires `due/`, and the edit grammar does not accept `due/NONE` or priority changes. Do not advertise those APIs as commands. Unknown IDs and rejected fields leave candidates, counters and plans unchanged; assertions supplement rather than replace runtime validation.

Design considerations

This is a present-day comparison of observable tradeoffs, not a claim about unrecorded historical deliberations. Mutating live tasks in place would allocate less, but require explicit rollback after saving failed. Immutable candidates make the save/publication boundary testable at the cost of list copies. Using a stored completion flag could represent another state independently of work, but would require reconciling inconsistent flag/estimate combinations. Here zero is the single completion authority. Rejecting excess credit exposes an inaccurate report instead of silently losing its meaning. The design intentionally stores neither an original-work baseline nor a timer/history, so it cannot report a whole-task completion percentage or reconstruct earlier estimates.

Dated commitments and save-before-publication dispatch

Students' fixed time changes in irregular ways: a lecture series ends mid-semester, one tutorial is cancelled, or a single afternoon becomes unavailable. This enhancement lets one `Commitment` record represent weekly, date-bounded and one-time time, checks conflicts against actual calendar dates, and guarantees that a rejected or unsaved change never reaches the live state.

Walkthrough: editing a commitment

Suppose C1 is `Lecture`, Monday 10:00–11:00 from 2026-11-02 until 2026-11-30, and the student enters `commitment edit C1 start/1030`.

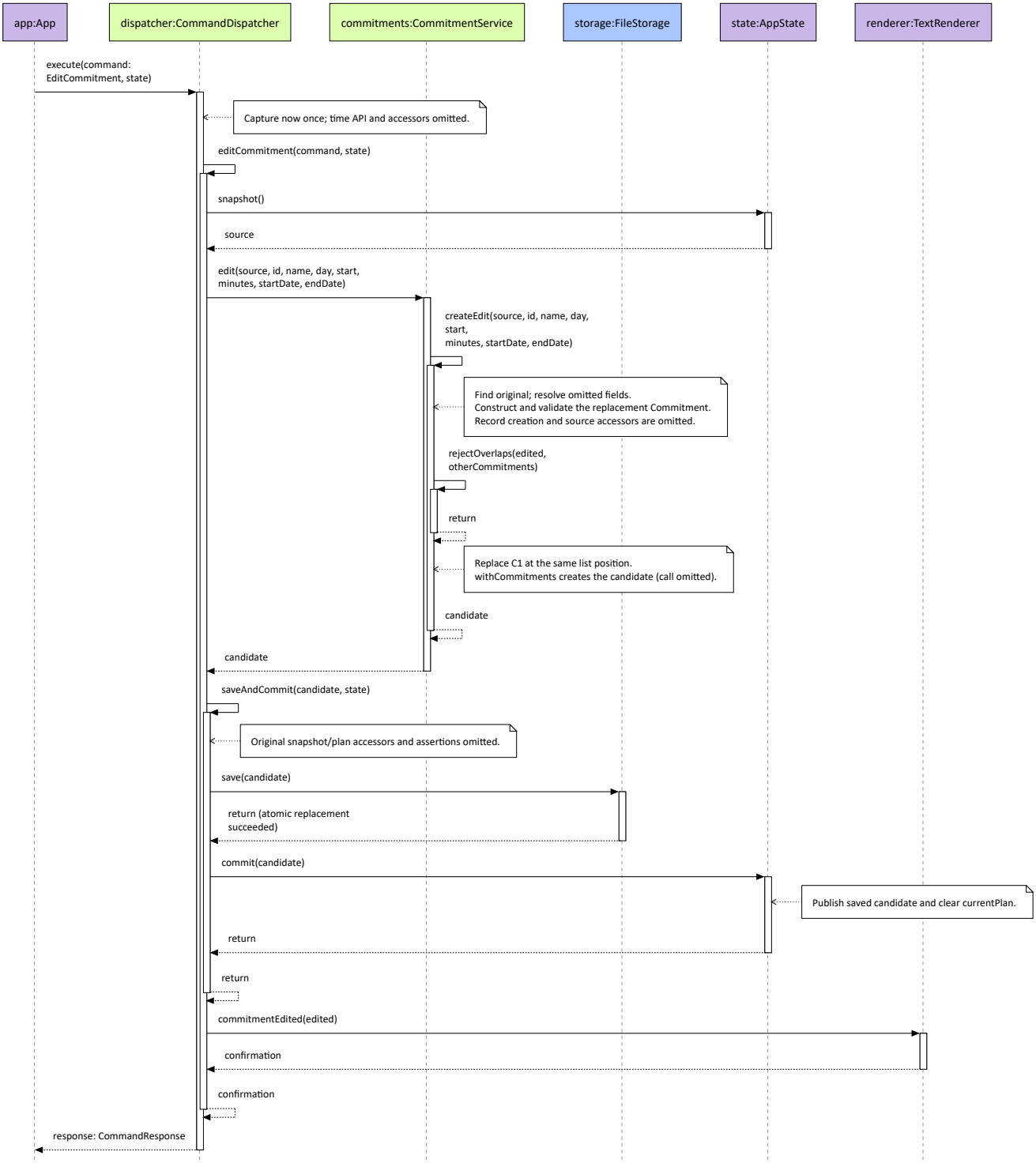


Figure 10. Successful commitment edit, saved before publication.

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Figure 10 shows a successful edit. Nested activations distinguish `execute`, `editCommitment` and `saveAndCommit`, and the service's `createEdit` and `rejectOverlaps`. Time capture, record construction and most value accessors are omitted; rejection paths are explained after the walkthrough.

1. Nickhil's parser produces `Command.EditCommitment(1, null, null, 10:30, null, null, null)`. `null` means "keep the stored value".
2. `CommandDispatcher.execute` captures `LocalDateTime.now(clock)` once, then `editCommitment` passes the live snapshot to `CommitmentService.edit`.

3. The service rejects a nonpositive ID or an edit with no fields. It then finds C1, fills each omitted field from the original, and constructs the replacement `Commitment`. The constructor checks alignment, duration and dates.
4. `rejectOverlaps` compares the replacement with every other commitment. C1 itself is excluded, so keeping or shifting its own time is never a self-conflict.
5. The service returns a new `Snapshot` with C1 replaced at its original list position. Tasks, study hours, policy, recurrence and all three ID counters are unchanged.
6. The dispatcher calls `saveAndCommit`. `Storage.save(candidate)` runs first; only if it returns does `AppState.commit(candidate)` publish the change and clear the current plan. `TextRenderer.commitmentEdited` then formats `Edited C1: Lecture | Monday 10:30 - 11:30 (60 min) | 2026-11-02 to 2026-11-30`.

If step 3 or 4 throws `ValidationException`, nothing was saved and `App` prints the message. If `save` throws `StorageException`, the dispatcher rethrows it before `commit`, so the live snapshot, current plan and saved file stay as they were and the same command can be retried. Add, skip and delete follow the same pattern; they differ only in the service call and which record the confirmation names.

Recurring commitment addition

`CommitmentService.add` creates an immutable candidate snapshot. It reuses `Commitment` for name, weekday, duration and interval validation, checks overlaps against the full weekly intervals (including outside study hours), and reports every conflicting commitment's ID and name in start-time order, with numeric ID as a tie-breaker. The private overlap check collects matches before rejecting; it never reorders the source snapshot. Single-conflict wording is unchanged, and multiple conflicts are separated by semicolons. Adjacent intervals and duplicate names at nonoverlapping times are allowed.

`CommitmentService.add` uses the next C ID and advances only that counter. IDs stop at `2147483646`; the exhausted next counter is `2147483647`.

Commitment editing

`CommitmentService.edit` accepts an existing numeric ID and nullable replacement fields: name, weekday, start and boxed duration. Null means omitted; a blank name is not an omission. It rejects empty edits and unknown/nonpositive IDs, resolves omissions from the original record, and constructs one validated replacement. The existing model and overlap helper remain the validation authorities. Only the target ID is excluded from conflict checks. Stored position, other records, tasks and both counters are preserved, even when the next ID is exhausted. No delete/add intermediate state or storage-format change is required. The edit candidate explicitly carries the source study-hours settings; using the legacy four-argument snapshot constructor here would silently restore defaults. A regression test round-trips both weekday and dated settings after editing. UI-014 also checks settings survive an edit and a fresh process restart.

`Command.EditCommitment` transports partial values. The parser accepts subsets in `n/`, `day/`, `start/`, `d/` order with the existing strict weekday/time/integer rules. It rejects repeated, reordered and unknown fields. Domain validation, including empty-edit rejection, remains in the service. The dispatcher resolves the replacement before the existing save-and-commit transaction and reports it through `TextRenderer.commitmentEdited`. Even identical-value valid edits save and invalidate the saved plan, consistently with successful mutations.

Recurring commitment date bounds

`Commitment` optionally stores a paired inclusive `startDate` and `endDate`. Both null means an unbounded legacy series. A bounded range must be ordered and contain at least one occurrence of the commitment weekday. `occursOn` is the single calendar-membership check used by allocation and schedule projection. Overlap requires intersecting times, the same weekday and at least one shared occurrence; therefore separate semesters may safely reuse a weekly time slot.

Addition and editing accept the ordered optional suffix `from/YYYY-MM-DD until/YYYY-MM-DD`. The parser requires both fields and strict four-digit dates. Omitted edit bounds preserve the stored range. Success and list output show inclusive dates only for bounded series, retaining existing text for unbounded records. Version 3 storage writes two endpoint fields on every commitment row and uses `-|-` for unbounded rows in the same file. Versions 1 and 2 remain readable, and stores without recurrence or bounded/skipped commitments use version 2 on new saves.

Skipping one recurring occurrence

`Commitment` stores an immutable set of skipped calendar dates. `recursOn` checks the weekday and optional inclusive bounds without exceptions, while `occursOn` additionally excludes skipped dates. The constructor rejects a skipped date outside the series or on another weekday. Editing retains the set, so an edit that would move an existing exception outside the resulting series is rejected instead of silently losing it.

`CommitmentService.skip` resolves the stable ID, rejects non-occurrence and duplicate dates, and replaces the record at its original stored position. The candidate preserves tasks, study hours and both ID counters. The dispatcher saves the candidate before publishing it or returning `Skipped CNUMBER on DATE: NAME`; therefore save failure leaves the live snapshot, current plan and saved data unchanged. The command grammar is `commitment skip CNUMBER on/YYYY-MM-DD`.

Overlap checks search the finite range whenever either series is bounded and require an unskipped shared occurrence. Two unbounded series always share a future occurrence because each stores only finitely many skipped dates. This allows another bounded commitment to use a cancelled date without weakening conflict checks on the remaining dates. The planner and schedule projection already use `occursOn`, so they require no separate exception logic.

One-time commitments

A one-time commitment is a `Commitment` whose inclusive `startDate` equals its `endDate`; its `day` is that date's weekday. `Commitment.isOneTime()` exposes the distinction. No record component, service operation, command type or storage version was added: the design reuses the issue #64 date-range model, so `occursOn` already blocks only that date in `EarliestDeadlinePlanner` and `ScheduleService`, and `overlaps` already compares the actual occurrence with recurring series (including skipped dates) and other one-time commitments. Version 3 rows such as `COMMITMENT|1|FRIDAY|14:00|60|2026-10-30|2026-10-30|NAME` store it; skipped dates require at least version 4 and recurrence requires version 5. Older stores remain readable unchanged.

The alternative of an explicit kind flag would need a new record component, constructor overloads and another format extension while behaving identically. The trade-off is that a weekly series entered with the same `from/` and `until/` date is shown as one-time, which matches its single occurrence. The packaged UI-017 expectation for that input was updated accordingly.

`CommandParser` accepts `on/YYYY-MM-DD` in the `day/` position of `commitment add` and `commitment edit`, and maps it to an ordinary `AddCommitment` or `EditCommitment` with the derived weekday and equal dates. Combining `on/` with `from/` / `until/` is a parse error; `day/` and `on/` are mutually exclusive in the grammar. Existing save-before-publish dispatch and ID allocation therefore apply unchanged: rejected input consumes no ID.

`CommitmentService` adds the domain rules that differ from weekly series. `skip` rejects one-time commitments, and the constructor rejects a one-time record whose only date is skipped, so corrupt saved data cannot hold a commitment that never occurs. `edit` rejects `day/` alone on a one-time commitment because its weekday follows its date, and discards skipped dates when an edit produces a one-time result. `TextRenderer` shows `One-time on DATE (Weekday) HH:mm - HH:mm (N min)` in add/edit confirmations, and `commitments` lists one-time records in a separate section ordered by date after the weekly groups, preserving the service's start/ID order within a date.

Commitment deletion and listing

`CommitmentService.delete` removes an entire weekly series by its stable numeric ID, returning an immutable candidate while preserving tasks, other commitments, their stored order and both ID counters. Nonpositive and unknown IDs raise `ValidationException`. Deleting the last series does not reset the counter. `list` returns an immutable Monday-to-Sunday view sorted by start time and then numeric ID, without reordering the snapshot or renumbering records. Valid commitments may share a weekday and start time when they have no shared, unskipped occurrence, for example in separate semester ranges. The numeric-ID tie-breaker gives those entries deterministic ordering.

Defensive checks distinguish invalid user IDs from programmer errors (null snapshots). Runtime validation works independently of Java assertions.

Unavailable study periods

A student marks an unavailable period using the existing one-time commitment command: `commitment add n/Unavailable on/YYYY-MM-DD start/HHmm d/MINUTES`. The name is an ordinary user description, not a reserved keyword. Equal inclusive date bounds represent the single unavailable occurrence, and `Commitment.occursOn(date)` supplies the same date membership check to the allocator and schedule projection. There is no separate availability model, new command or storage format for this workflow.

A period may overlap previously allocated task sessions: commitment overlap validation checks fixed commitments, not temporary task allocations. The existing save-before-publication dispatcher invalidates the old plan after a successful addition. Regenerating allocates tasks around the blocked interval, preserves task estimates and reports any work that no longer fits before its deadline. The schedule displays the period as `[BUSY]` with its ordinary commitment ID. Deleting that ID saves and invalidates again; replanning can use the released interval. Previously returned immutable plan objects stay frozen.

The existing half-hour alignment, positive duration, same-day interval and fixed-commitment overlap rules apply. Adjacent task sessions can end at the block's start or begin at its end. Blocking every available slot produces an unallocated-work result rather than scheduling after the task's deadline. Hours overrides continue to control the enclosing study window.

Command dispatch and live-state transactions

`CommandDispatcher.execute` is an exhaustive `switch` over the sealed `Command` type, so adding a record without a route is a compile error. It captures `LocalDateTime.now(clock)` exactly once per command and passes that value to every time-dependent step, including `schedule today`, so one command never sees two different "now" values.

Commands fall into four groups:

Group	Examples	Saves?	Effect on the current plan
Read-only	`help`, lists, `hours DATE`, `policy show`, `schedule`, `exit`	No	Kept
Source mutation	task, commitment, hours, policy and recurring-template changes	`Storage.save(candidate)`	Cleared after a successful save
Unchanged mutation	zero-credit progress, an unchanged task edit or policy	No	Kept
Planning	`plan`	`Storage.savePlan(plan)` once	Replaced

Every source mutation goes through the private `saveAndCommit(candidate, state)`. It calls `Storage.save` and, only on success, `AppState.commit`. Assertions check that the state is untouched before saving and that the plan is empty afterwards. Handlers that may return the original snapshot unchanged compare it with the candidate by reference and skip the save, so a no-op neither writes the file nor discards a valid plan. Commitment edits are not treated as no-ops: an edit with identical values still saves and clears the plan, consistent with other successful edits.

`generatePlan` is a single transaction across Gabriel's and Ernest's components:

1. `RecurringTaskService.materialize(source, now)` returns the same snapshot, or a candidate with newly released weekly occurrences and advanced cursors.
2. `Planner.generate(candidate, now)` allocates sessions. The dispatcher rejects a plan whose `source()` differs from the candidate.
3. `TextRenderer.plan(plan)` prepares the output before anything is saved.
4. `Storage.savePlan(plan)` writes the source and frozen result together.
5. Only then is the live state updated: `commit(candidate)` if materialization changed the source, followed by `replacePlan(plan)`.

There is deliberately no separate snapshot save followed by a plan save. If step 4 fails, the previous snapshot, plan and file all remain current.

`showSchedule` checks for a current plan before projecting a date, so "no current plan" takes precedence over date-range errors. Domain exceptions remain their owners' validation; the dispatcher neither catches nor rewords them.

Design considerations

Aspect: when to publish a change

- **Alternative 1 (current): save, then publish.** Services build immutable candidates; the dispatcher saves before calling `AppState.commit`.
  - Pros: a failed save leaves memory, disk and plan consistent, so a retry reuses the same ID. Services stay free of I/O and are easy to test.
  - Cons: each mutation builds a new `Snapshot`, and every handler must call `saveAndCommit` rather than mutating state directly.
- **Alternative 2: mutate live state, then save (rolling back on failure).**
  - Pros: less copying.
  - Cons: a failed save would need a hand-written undo for every operation. A missed rollback would show the student data that is not on disk and consume an ID that the file never recorded.

Aspect: representing one-time and dated commitments

- **Alternative 1 (current): one record, with equal dates meaning one-time.**
  - Pros: no new type, command or storage version. The planner and schedule use the same `occursOn` and `overlaps` for every shape.
  - Cons: a weekly series entered with the same `from/` and `until/` date is shown as one-time. That is equivalent, because it has exactly one occurrence.
- **Alternative 2: separate weekly and one-time classes, or a kind flag.**
  - Cons: every consumer would need type checks, the storage format would need another field, and overlap logic would need pairwise variants.

Aspect: where conflicts are checked

Overlaps are checked in `CommitmentService` against all other commitments, including outside study hours, rather than in the planner. Planning therefore never has to choose between two clashing fixed commitments. A task session may still be overlapped by a new commitment; the next `plan` simply allocates around it.

Multi-date unavailable periods

`AddContinuousBlock` and `AddDailyBlocks` reuse the same one-time commitment representation as a single unavailable date. `CommitmentService` expands the request and validates every generated period before returning one candidate. Continuous mode treats its end as exclusive and splits at midnight; daily mode uses one same-day window on every date in its inclusive range. Generated records are appended in date and allocated-ID order.

`CommandDispatcher` passes the original snapshot to the matching service, takes only the appended commitments for `TextRenderer.unavailablePeriodsAdded`, and prepares that complete confirmation before persistence. It then calls the normal `saveAndCommit` boundary exactly once. A successful save publishes the complete batch and clears the old plan once. Validation, rendering or save failure leaves the original commitments, counters, plan and disk unchanged, so a retry receives the same first C ID. Individual generated commitments remain ordinary one-time records and can be edited or deleted by their C ID.

UI-022 through UI-027 in the active fixture cover continuous and daily output, restart, independent deletion, invalid syntax and values, later-segment rollback, exclusive midnight, adjacency and leap day. Focused dispatcher tests use controlled storage to establish one-save ordering, failed-save retry, ID exhaustion and preservation of unrelated snapshot data.

Slot allocation under study hours/policy and recurrence

Frozen allocation validation

`Planner.validate` rejects unknown/completed task references, duplicate remainder rows, identity/deadline/reason mismatches and missing or excessive work. Long sums require scheduled `workMinutes` plus one optional remainder to equal each source task's exact remaining work. Padding is permitted, so a 45-minute task can occupy 60 minutes. Feasibility does not require greedy or optimal placement.

Task and break intervals must fit effective hours, the frozen cutoff and horizon. Task intervals must also fit their current deadline and original occurrence release. Half-open intervals allow exact adjacency; task/task, task/break, break/break and allocation/occurring-commitment overlaps are rejected. Commitment membership uses `occursOn`, including bounds and skipped dates. Overdue and beyond-horizon work remains fully unallocated with its source reason; undated work can only report insufficient capacity. Validation checks break bounds and conflicts without replaying policy placement. Legacy/fallback-policy breaks are accepted even without a stored policy row. Null plans throw `NullPointerException`; inconsistent allocations throw `ValidationException`. No I/O, clock reads, recurrence materialization or source mutation occurs.

Exact work and slot capacity

Exact work and slot capacity are distinct: `StudySession.workMinutes` carries positive exact work while `minutes()` remains the reserved interval capacity. The original three-argument constructor defaults work to capacity. `UnallocatedWork` carries an optional deadline, exact `int minutes` and rounded `long capacityMinutes`; its original concrete-deadline constructor remains. Demand is `30 * ceil(R / 30)` using long arithmetic. Allocation consumes `min(remaining, 30)` exact work per slot, never subtracting below zero; merged sessions retain that exact sum. For R=45, 60 allocated capacity means 45 planned work; 30 capacity leaves 15 exact work and 30 unallocated capacity. Source estimates are never decremented. Totals must use long streams. Only the final session for a task can contain padding, and zero-work sessions are not created.

Continuous work and fixed breaks

The console supports `policy set max/MINUTES break/MINUTES`, `policy show` and `policy reset`. `StudyPolicy` validates positive multiples of 30 minutes. `Snapshot.studyPolicy()` is an explicit optional: absent means unrestricted planning, preserving legacy behavior without inventing numeric defaults. The no-argument planner reads the stored policy on each generation. The explicit `EarliestDeadlinePlanner(StudyPolicy)` constructor remains available as a fallback for API clients whose snapshot has no policy; a stored policy takes precedence.

Allocation now walks slots chronologically, choosing dated unfinished tasks by deadline, priority and numeric ID, then undated tasks by priority and ID. This retains deadline/ID order for equal-priority dated tasks while allowing one work counter to cover different tasks and commitments. Fixed commitments count even outside study windows and before the rounded generation cutoff. The counter starts at midnight on the generation date; activity before that date and past task study are not reconstructed. Across future dates, calendar midnight does not itself reset the counter: actual unoccupied time does.

Tasks may immediately follow commitments while the combined work total is below the limit. At the limit, another task slot must wait for the configured consecutive rest. Short idle gaps retain accumulated work; any new work interrupts partial rest. Fixed commitments are never moved or shortened, even when they push the combined work total above the limit. The allocator does not reserve rest in anticipation of a future commitment. After an overrun of any size, one configured break suffices; proportional break scaling is deferred. Natural idle time, including time outside study hours, can satisfy all or part of the required rest.

`Plan.breaks()` contains chronological immutable `ScheduleEntry` rows with `SlotKind.BREAK` and empty reference/name fields. Its original five-argument constructor remains available and supplies an empty break list. Relationships between sessions, breaks and commitments remain the planner's responsibility. Only eligible slots withheld from pending, deadline-eligible tasks are marked as breaks; natural rest and trailing idle time remain ordinary free/unplanned time. An interrupted break can therefore appear as separate partial rows, and a break may end at a deadline even if the task ultimately remains unallocated. `ScheduleService` projects frozen break rows without recomputing the policy.

`CommandParser` transports integer minutes without constructing domain settings. The dispatcher creates `StudyPolicy`, obtains a policy-only candidate and formats the response before saving. Changed settings are saved before publication and plan invalidation. Equal sets and an already-unrestricted reset return the same snapshot, skip saving and preserve the current plan. Failures preserve both the stored settings and saved plan. `policy show` is read-only. All task, commitment, hours and recurrence candidates preserve the optional setting.

Configurable study hours

`StudyWindow` validates positive same-day intervals on half-hour boundaries. `StudyHours` holds an immutable map keyed by full lowercase weekday names or canonical ISO dates. Resolution is date override, then weekday, then `StudyWindow.DEFAULT` (08:00–22:00). Resets remove only the selected key. Overnight windows and 24:00 endpoints are unsupported; the latest endpoint is 23:30.

`Snapshot` now includes `studyHours`; its original four-argument constructor remains available and supplies defaults. Task and commitment services carry the settings into all candidates. `SetHours`, `ShowHours` and `ResetHours` commands use the agreed `hours TARGET /start HH:mm /end HH:mm` grammar, full lowercase weekday names and strict ISO dates. The dispatcher saves a changed snapshot before publication and plan invalidation; failed saves retain the old plan. Reset dispatch uses a separate confirmation from setting hours. The renderer identifies whether the restored window comes from a recurring weekday setting or the default and prints its endpoints. Dispatcher tests cover both sources and resetting an unset date; UI-015 checks the exact reset confirmations.

Both slot allocation and schedule projection resolve the window from the snapshot. `TimeRules.eligibleStart` has a window-aware overload; the original overload retains default behavior. `StudySession` and `ScheduleEntry` validate same-day, positive, aligned intervals; the planner enforces configured hours. `ScheduleView` carries and

validates its effective window, retaining the original four-argument constructor for default-window callers. Plans freeze their source settings, so later time or unrelated snapshots do not alter projections.

Weekly recurring tasks

`RecurringTaskService` integrates weekly recurrence with the exact-work, optional-deadline, priority and lifecycle model. `RecurringTask` stores an independent positive R ID, normalized name, first deadline, optional inclusive final date, positive half-hour estimate and persisted last-processed deadline. `TaskOccurrence` retains the template ID and original deadline; release is always one week before that original deadline. Ordinary task additions accept positive whole minutes, including 45; completed records retain zero internally. Template estimates remain positive multiples of 30. Generated tasks subsequently support exact remaining revisions and domain progress/completion/reopening while retaining provenance. The four-argument Task and four/five-argument Snapshot constructors remain usable for shipped callers and legacy data, defaulting recurrence to empty.

`task recur add`, `task recur list` and `task recur delete` are parsed, dispatched and rendered at the console. The first deadline establishes weekday/time; only `every/WEEKLY` is supported. Creation materializes all future due dates through the rolling horizon (including its boundary), subject to an optional last date. Future templates beyond the horizon are retained without premature tasks. Materialization skips missed deadlines arithmetically rather than backfilling; the saved cursor also covers skipped dates. Pending tasks get T IDs in deadline, then numeric-template-ID order. The complete batch must fit before any ID is allocated. Replanning, a backwards clock change, task deletion and restarting never rewind the cursor or recreate a processed occurrence.

Snapshot validates distinct logical occurrences, cursor coverage and the separate R counter, including retained provenance after a template is deleted. Task edits, remaining revisions, deletions, commitment changes and study-hours changes retain the recurrence definitions/counters. Editing one generated task does not edit its template or release. Template deletion stops only new generation and retains all already-generated tasks. The allocator checks the original release against slot start while retaining current study hours, dated/skipped commitments and continuous-work policy. No second allocation algorithm is introduced.

`plan` first materializes a candidate and generates its Plan, validates the source association and formats the result before publication. It atomically saves the complete candidate and frozen plan together through `Storage.savePlan`, then commits any changed source and publishes the plan. This avoids an intermediate snapshot-only save that could discard the previous plan if the second save failed. Planner/null/stale-result/rendering/save failure preserves the previous snapshot, cursors, IDs and plan. Every successful `plan` saves its result, including when materialization makes no source changes.

Atomic persistence and restoring frozen plans

Consider a student who generates a plan, closes the app and returns later. Saving only task estimates would require a new allocation using a later clock time. Saving only session rows would lose the source needed to interpret task IDs, study windows and recurring occurrences. `savePlan(plan)` instead stores the complete source and frozen result in one file. On restoration, the schedule keeps its original sessions and generation time; reading it does not credit progress.

Atomic snapshot and plan saves

Both save operations use `FileStorage.writeAtomically`:

1. `save(snapshot)` calls `StateCodec.encode`; `savePlan(plan)` calls `PlanCodec.encode`, which validates the allocation with `Planner.validate` before asking `StateCodec.encodeForPlan` for the full source. Encoding completes before any filesystem preparation. Null arguments are rejected before filesystem changes.
2. Create missing parent directories and a distinct sibling temporary file. Keeping both paths in one directory avoids a cross-filesystem move.

3. Write the complete text using UTF-8, then call `Files.move` with `ATOMIC_MOVE` and `REPLACE_EXISTING`. There is no ordinary-move fallback.
4. Return only after replacement succeeds. On a filesystem failure, attempt to remove the temporary file and throw `StorageException`; attach a cleanup failure as suppressed information instead of replacing the original error.

The public constructor installs `FileStorage.FileOperations`, whose creation, write, atomic-replacement and deletion methods call the real `Files` API. A package-private constructor accepts controlled operations for tests. Directory creation and all reads remain direct filesystem calls. The seam does not change the public interface, encoding, replacement options or error handling, and construction still performs no I/O. If temporary creation fails before a path is returned, there is no known temporary file to clean up. After write/move failure, cleanup is attempted once; a failed cleanup may leave an abandoned `.scheduleflow-*.tmp` sibling while retaining the original target.

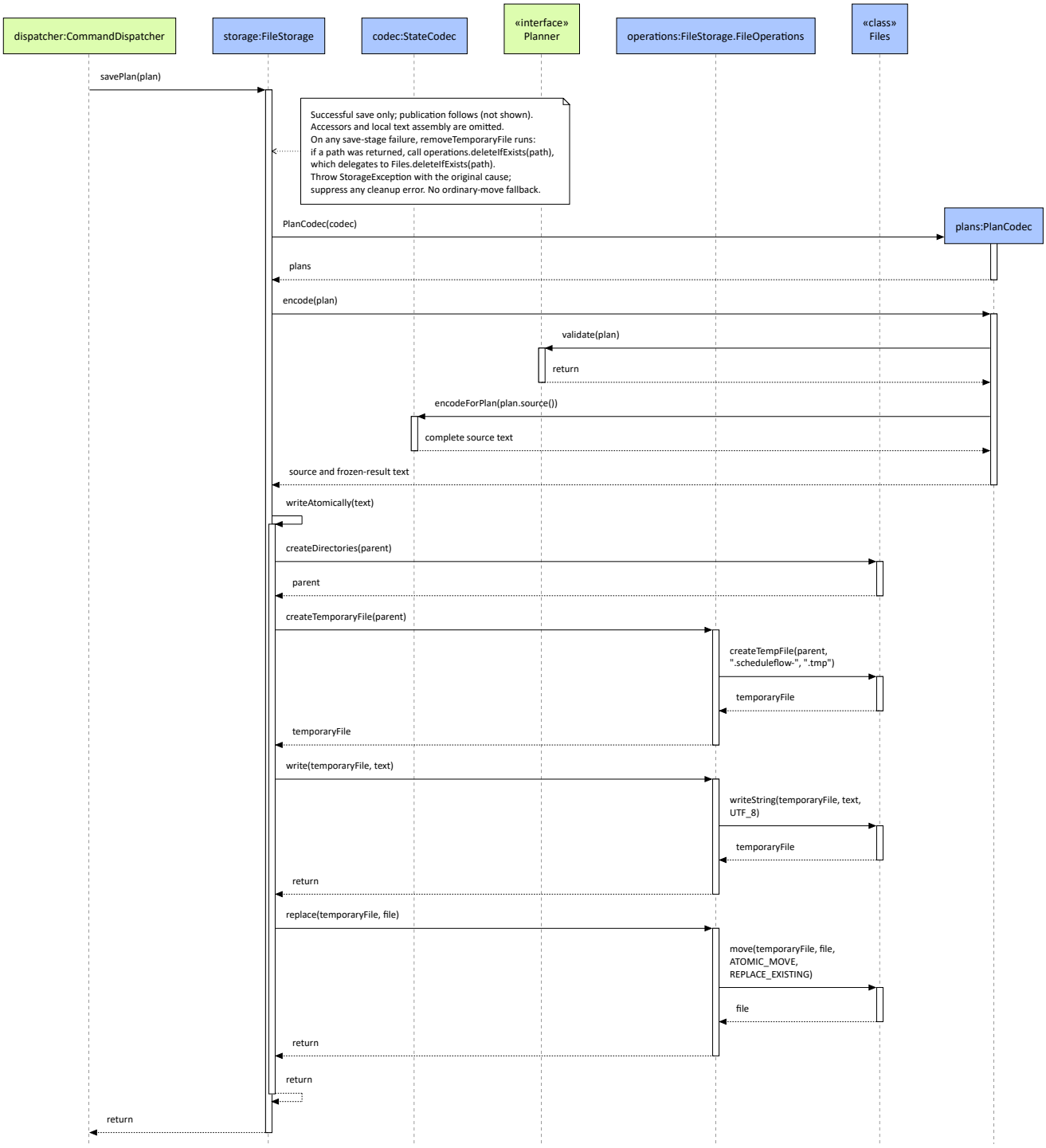


Figure 11. Successful atomic save of a frozen plan.

PlantUML source · PNG. The diagram abbreviates static filesystem calls and follows a successful atomic replacement. `Files` is a class participant, and `PlanCodec` is created during `savePlan`, before `encode` is called. The operations collaborator forwards normal calls to `Files`; its note traces failure cleanup through the same collaborator. See command transactions for live publication: the dispatcher prepares recurrence/allocation/output, calls `savePlan` once, then publishes the saved source and plan. It does not save the source separately first. A rejected save leaves the previous live snapshot and plan in use, allowing retry without consuming IDs or advancing template cursors.

A successful `save(snapshot)` deliberately writes no plan. It removes a stale persisted schedule after a source mutation. Read operations and service-defined no-ops do not call it. This distinction lets an unchanged policy or zero credited progress preserve the current schedule while a changed estimate invalidates it.

Startup restoration

`FileStorage.load()` reads the complete file. Only a target that is definitely absent returns `Snapshot.empty()`; access errors, a directory target, malformed UTF-8 and invalid records are failures. A broken symbolic link must not silently become an empty store. Loading itself creates no directories and rewrites no legacy data.

`StateCodec.decode` validates headers/records and constructs domain values and the complete `Snapshot`. Saved overdue tasks are valid: restoration does not repeat the time-dependent admission rule for adding a new task. For a plan file, decoding also checks metadata, counted rows, task references, matching remainder names/deadlines, interval bounds and result ordering. `Planner.validate` then checks feasibility and work conservation against the frozen source/time. Encoding uses the same boundary before any directory or temporary file is created. Neither path reruns the allocator or materializes occurrences.

`App.run()` loads the snapshot and calls `loadPlan(source)` before the first prompt. `loadPlan` rereads the file and accepts only an equal embedded source; a change between reads is rejected when it changes that source. Snapshot-only files return no plan. Invalid source or plan data stops startup instead of returning partial records. This is not a file lock or concurrent-writer protocol.

File formats and compatibility

The current decoder accepts the following layouts; the encoder selects the smallest snapshot layout required by its records, or a plan layout explicitly. Version numbers describe saved data, not application release numbers.

Header version	Data represented / current write rule
1	Read-only legacy layout: `TASK`, weekly commitments and T/C counters; default hours, NORMAL priority and no policy.
2	Minimum new snapshot: `TASK2` exact remaining minutes (including zero), optional deadline/priority, `HOURS`, and T/C counters. Legacy `TASK` rows still require positive half-hour work.
3 / 4	Add commitment date bounds / skipped dates respectively; selected when those fields are needed without recurrence or policy.
5	Snapshot with templates, occurrence provenance or an allocated R counter; includes all three counters and the widest commitment layout.
7	Full version 5 source plus exactly one valid `POLICY` row. Resetting policy permits a lower snapshot version when appropriate.
6 / 8	Frozen plan plus full version 5 / 7 source respectively, including exact credited work, optional remainder deadlines, long rounded capacities and breaks.
Legacy 5 plan	Read-only earlier prototype, distinguished by two `NEXT` counters; source uses version 4 and sessions credit their full interval. New plan writes use 6 or 8.

`NEXT` counters survive deletion. `RECURRING` rows retain generation cursors and `OCCURRENCE` rows retain original template/deadline provenance, preventing lost identity across restarts. Names are the final field so literal pipes remain part of the name. Task/commitment list order is preserved; hours targets and skipped dates are encoded in sorted order. Encoding uses LF with a final newline; decoding also accepts CRLF/CR and an omitted final newline. Session rows preserve exact work separately from reserved half-hour capacity: a 45-minute task can occupy 60 minutes without becoming a 60-minute estimate.

Persistence tradeoffs and limits

One file keeps source/result replacement indivisible and remains inspectable, but each save rewrites the complete state. An alternative database could support coordinated updates and schema migrations at greater setup/dependency cost. This is a design analysis, not a claim about historical team deliberations.

Atomic replacement requires filesystem/provider support, and only one running instance may write a given file. It provides neither writer coordination nor a power-loss durability guarantee: the implementation uses `Files.writeString`, not `FileChannel.force(true)`. Importing a different wire format requires explicit conversion and verification against these codecs. Back up before changing application versions; older executables may reject newer layouts.

Documentation, logging, testing and dev-ops

Review both guides with each implementation, following AGENTS.md. Use isolated saved data and fixed clocks where exact dates matter.

Build, packaging and console verification

From the repository root, verify both the shell and Gradle JVM are Java 25:

```
java -version
.\gradlew.bat --version
.\gradlew.bat build
node --test test/ui-runner.test.mjs
node .agents/skills/test-ui/scripts/run-ui-tests.mjs
node --test test/release.test.mjs
```

On Unix use `./gradlew`. If necessary set `GRADLE_USER_HOME` to a writable cache. `build` runs JUnit and Checkstyle and creates `build/libs/scheduleflow.jar` with entry point `scheduleflow.Main`. Standalone launches require `java -ea -jar scheduleflow.jar` to enable assertions; Gradle `test`, `run` and distribution scripts enable them already. `startScripts` and normal distributions use the same Shadow JAR; test dependencies are not shipped.

For focused shell checks, run `test --tests scheduleflow.MainTest --tests scheduleflow.cli.AppTest --tests scheduleflow.cli.AppStateTest --tests scheduleflow.cli.ConsoleUiTest --tests scheduleflow.cli.TextRendererTest` with the wrapper. The UI plan explains isolated data, per-command comparisons and scenario readiness. The UI runner rebuilds the current JAR and tests without assertions; `release.test.mjs` replays maintained cases with `-ea` and extracts/launches the normal distribution ZIP. It checks both scripts' assertion defaults but executes only the current OS's script. Preserve the printed `build/ui-transcripts/` paths before running `clean`.

The CI workflow retains its Ubuntu, macOS and Windows matrix and Java 25 setup. It installs Node 22, prints the Java, wrapper and Node versions, then runs `build`, runner self-tests, active UI cases and the release replay in that order. The wrapper is selected for each runner OS; packaging finishes before distribution acceptance, and the two process suites never regenerate the build directory concurrently. Any failed check fails its job. Independent matrix jobs continue to collect evidence when another OS fails. An always-run artifact step retains only JUnit/Checkstyle reports and console/ release transcripts, excluding saved data and diagnostic logs. Workflow changes and local Windows results do not establish hosted or other-platform success.

Diagnostics

Default `Main` launches silence `scheduleflow.cli` unless logging was explicitly configured. Embedded `App` instances respect the caller's configuration. Stage logs omit raw commands/output, but failure records retain exception details; keep diagnostic files private and outside version control.

For opt-in file diagnostics, put this `logging.properties` in a disposable launch folder and run `java -ea -Djava.util.logging.config.file=logging.properties -jar scheduleflow.jar`:

```
handlers=java.util.logging.FileHandler
.level=INFO
scheduleflow.level=FINE
java.util.logging.FileHandler.level=FINE
java.util.logging.FileHandler.pattern=scheduleflow-%u.log
java.util.logging.FileHandler.encoding=UTF-8
java.util.logging.FileHandler.formatter=java.util.logging.SimpleFormatter
```

`FINE` records stages, `INFO` session start/normal termination, `WARNING` recoverable rejections and `SEVERE` startup/input/output failures. To retain warnings and failures only, set both `scheduleflow.level` and the handler level to `WARNING`. Constructors perform no I/O or logging configuration.

Diagram and guide maintenance

Editable PlantUML sources and the shared `course-style.puml` are in `docs/diagrams`. Each figure has a numbered caption and links to its source and PNG; committed SVGs in `docs/images` are embedded in this guide.

Review notation against the course's Modeling chapter and Week 11 DG review guidance. Check depicted members, retained references and call order against the current Java source and its tests. Keep each view focused; identify omitted helper calls in attached notes or the surrounding explanation. Although the textbook permits unattached notes, this guide connects every note with a dotted arrow to its subject.

The DG review (#118) identifies course notation, missing activations and unclear notes as concerns. The figures retain the existing PlantUML workflow and share a flat pastel palette for responsibilities. Exports were checked with PlantUML **1.2025.4**, Graphviz **2.44.1**, Java **25.0.4.1**, and installed **Calibri** (no font fallback). The SVG-to-PNG step uses Playwright **1.62.1** and Chrome **156.0.8078.49**. The shared style selects strict UML notation, textual visibility and rectangular pastel boxes. Fills, spacing and curved structural routes are presentation choices; the arrows, visibility, activations and other UML symbols follow course notation. SVGs use logical vector dimensions; PNGs are exported at 2x resolution, with a white background and preserved aspect ratio.

With Java 25, Graphviz, Node.js 22+, Playwright and a local PlantUML JAR, render all figures from the repository root. Install documentation-only Node dependencies under ignored `build/uml-tools` if they are not already available:

```
npm install --prefix build/uml-tools --no-save --package-lock=false playwright
$env:NODE_PATH = Join-Path (Get-Location) 'build/uml-tools/node_modules'
node tools/render-uml.mjs --plantuml "PATH_TO_PLANTUML_JAR" --chrome "PATH_TO_CHROME"
```

Replace the two paths with installed PlantUML and Chrome locations. An existing Playwright-managed Chromium can be used by omitting `--chrome`. Use `--output DIR` to preview exports outside `docs/images`. The common include is a style file, not a standalone diagram. Captions belong in the guide so that titles cannot change export names. The Windows PlantUML distribution used here supplies Graphviz; other installations may require it separately.

PlantUML draws structural note arrows directly. Its sequence notes have no connector, so `tools/render-uml.mjs` adds dotted arrows to their named participants' active invocation bars (or lifelines), then derives the PNG from the same SVG. Sequence notes must use `note right of PARTICIPANT`; the exporter checks their count, text and targets

and reserves a visible gap for each arrow. Plain PlantUML exports alone omit these sequence note connectors.

Commit each edited source, shared style change and corresponding SVG/PNG exports together. Check displayed members and call order against the source and relevant JUnit tests. Inspect every rendered figure at full size: visibility and static underlines, endpoint roles/multiplicities, realization triangles, dotted note attachments, solid filled call heads, dashed returns, continuous/nested activations, guarded fragments, and constructor-time headers. Deliberate omissions are identified beside the figures; failure paths remain documented in prose.

Review both guides after implementation; mark unfinished operations explicitly. Before publication, the release maintainer must reconcile this guide with the verified release tag and artifact. Cross-platform runtime checks and filesystem provider fault injection need separate evidence; passing on Windows does not establish those results.

Task/parser validation and JUnit/documentation conventions

Run the owned tests with Java 25 from the repository root:

```
./gradlew test --tests 'scheduleflow.task.TaskServiceTest' --tests 'scheduleflow.task.TaskEditTest'
--tests 'scheduleflow.task.TaskRecurrenceTest' --tests 'scheduleflow.model.TaskTest' --tests
'scheduleflow.common.*' --tests 'scheduleflow.cli.CommandParserTest'
```

`TaskServiceTest` checks 0/50/100% credit, floor rounding, large products, no-op identity, completion/reopening, and captured-time deadline boundaries. TaskRecurrenceTest checks generated-task metadata preservation, including the complete lifecycle after template deletion with custom hours/policy and surrounding tasks; it does not replace Gabriel's generation tests. TaskEditTest and InputRulesTest cover transport/null rules and normalization. CommandParserTest covers all families' grammar and usage hints, not command execution. Explicit rejection tests keep domain-only optional-deadline, deadline-clearing and priority APIs outside the console grammar. Tests use explicit times, temporary data, and restored logger/locale settings; see test conventions. Link documentation claims to those tests rather than adding tests for prose.

Opt-in `FINE` diagnostics on `scheduleflow.task`, `scheduleflow.model.Task`, `scheduleflow.common.InputRules` and `scheduleflow.cli.CommandParser` omit raw names/command values. Use the existing file-handler configuration; these classes install no handlers. Public validation works without `-ea`; assertions check internal invariants. Lifecycle methods do not all emit stage logs. JUnit/documentation standards remain the authority for Javadoc, isolation and verification; builds alone are not manual acceptance.

Commitment/dispatch transaction verification

Three JUnit classes cover this component, all with a fixed `Clock` and in-memory or temporary storage, so no test reads the real time or the developer's data:

Test class	Tests	What it establishes
`CommitmentTest`	13	Record validation boundaries, `occursOn`/`recursOn`, adjacency versus overlap, shared-occurrence checks for bounded and skipped series.
`CommitmentServiceTest`	52	Each operation returns a candidate and leaves its source unchanged; conflict messages and ordering; edits keep position, ID and study hours; skip/one-time rules; counter exhaustion; unavailable-range expansion.
`CommandDispatcherTest`	46	Every route and a single clock read per command; save-before-publication for each mutation and unavailable batch; failed saves preserving state, plan, counters and retry IDs; no-op and read-only commands making no save; the one-`savePlan` planning transaction.

The dispatcher tests use an in-memory `FakeStorage` that records attempts and can fail on demand. For example, `execute_eachFailedSave_preservesStatePlanCountersAndRetryId` makes every mutating command fail once. It asserts that the live snapshot, plan and ID counters are unchanged, then retries and checks that the identical candidate is saved. `execute_planRejectedAfterMaterialization_preservesSourcePlanAndDisk` covers a plan rejected after recurrence materialization. Packaged console scenarios UI-014 (partial edits, rejection, plan invalidation and restart) and UI-017 (occurrence skips and restart) exercise the same rules through the JAR. UI-022 through UI-027 exercise both unavailable batch routes, rollback boundaries, restart persistence, independent deletion and calendar boundary cases.

Commitment defensive checks and diagnostics

Domain validation works without assertions; constructors perform no I/O or logging. Commitment diagnostics use `FINE` on `scheduleflow.commitment`, without names, schedules or IDs. Use the file-handler configuration in the console diagnostics section, selecting this logger namespace.

Fixed-time planning and recurrence verification

`PlannerTest` directly verifies alternative feasible frozen allocations, exact partial work/padding, legacy breaks, source preservation, accounting errors, overlap/adjacency, cutoff, overridden hours, deadlines, horizon, original release and commitment skips. Existing generator tests now exercise the same validator on every real result, retaining ordering, hours, policy and recurrence coverage.

`EarliestDeadlinePlannerTest`, `ScheduleServiceTest` and `RecurringTaskServiceTest` use fixed times to verify conservation of exact work, rounded capacity, horizon/cutoff boundaries, hours/policy, frozen projections, weekly releases and cursor behavior. Packaged runs use the live clock and cannot establish those exact-time allocation assertions.

Storage diagnostics and compatibility verification

`StateCodecTest` checks legacy/current formats, pipes/Unicode, exact work, bounds/skips, hours, policy and recurrence relationships. `PlanCodecTest` checks complete frozen results, malformed rows and distinct legacy layouts. `SnapshotTest` covers immutable copies, counters, overlaps and policy-preserving candidates; `FileStorageTest` uses temporary folders for byte-preserving reads, restarts, blocked parents, failed replacement and cleanup. Controlled `FileOperations` faults additionally exercise temporary creation failure, a partially written temporary file, `AtomicMoveNotSupportedException`, another replacement `IOException`, and cleanup failure after replacement failure. The write and unsupported-move cases cover both snapshot and plan saves through the shared transaction. Tests retain a real previous frozen plan, compare its bytes/source/result after failure, check cleanup/order and preserve the original cause. A cleanup error is suppressed on the `StorageException`; its intentionally retained sibling stays within JUnit's temporary folder. No ordinary-move fallback is attempted. Dispatcher/App tests exercise save-before- publication and matching-source restoration with fixed clocks and controlled storage. Their fake failures do not simulate every filesystem provider.

UI-031/UI-032 in the console test plan cover restored plans and invalidation. The manual recipes add backup/transfer and an interactive rejected-save check. These new faults are injected at real transaction boundaries, not evidence of unsupported moves or cleanup denial on an actual provider. Other operating systems, real permission denial, unsupported providers and power-loss behavior still need separate evidence; the implementation makes no power-loss guarantee.

`PlanCodecTest` rejects duplicate/missing/excess work, overlapping sessions, post-deadline work and pre-cutoff sessions, retaining validation causes through `StateCodec.decode`. Valid non-greedy plans and legacy/current formats round-trip. `FileStorageTest` verifies both load paths preserve impossible input bytes and invalid plans fail before filesystem changes or replacement of a valid plan.

Storage defensive checks and diagnostics

Runtime exceptions reject invalid data even with assertions disabled. Assertions check internal record-count and sibling-temporary-file invariants; use `-ea` for a packaged launch. `FileStorage` / `StateCodec` emit `FINE` stage diagnostics without paths, names, file contents or exception payloads in those log records. The application owns user-facing errors and session severity. Thrown exceptions may contain paths/details, so review logs before sharing them.

For storage-only logging, create `logging.properties` in a disposable launch folder (not the repository):

```
handlers=java.util.logging.FileHandler
.level=OFF
scheduleflow.storage.level=FINE
java.util.logging.FileHandler.level=FINE
java.util.logging.FileHandler.pattern=scheduleflow-storage-%u.log
java.util.logging.FileHandler.encoding=UTF-8
java.util.logging.FileHandler.formatter=java.util.logging.SimpleFormatter
```

Launch `java -ea -Djava.util.logging.config.file=logging.properties -jar scheduleflow.jar`. Default launches stay quiet; storage installs no global logger configuration.

Appendix A: Product Scope

Target user profile

University students balancing coursework deadlines, project work, classes and other fixed events, who are comfortable typing commands in a terminal. They need to revise estimates as work progresses and see when available study time is insufficient.

Value proposition and boundaries

ScheduleFlow turns remaining-work estimates, deadlines and busy intervals into a local study schedule. Stable IDs and saved settings let a student correct entries and continue later without re-entering data. Whole-plan unallocated work makes shortfalls visible instead of silently placing work past deadlines.

The CLI supports tasks and their retained completion/progress, weekly task templates, recurring/one-time commitments, configurable daily hours and an optional work/break policy. Planning uses half-hour capacity within a rolling calendar year; exact task work can be any positive whole-minute estimate. Schedules stay frozen until regenerated or invalidated by a successful change.

The application is a single-user, local/offline application with one running instance per data file. It has no calendar synchronization, GUI, notifications or automatic progress timer. Multi-date unavailable commands remain proposed console integration; see the User Guide for the supported command surface.

Appendix B: User Stories

These are product stories, not a redistribution of implementation credit. **Candidate** means supported by the inspected source, newer than published v1.0; **Planned** means not a working console feature. Feature owners should review their rows against the final release tag before publication.

Status	As a...	I want to...	So that I can...
Candidate	new user	launch locally and request help	learn the accepted commands without leaving the terminal.
Candidate	student	add and list coursework with deadlines and exact estimates	keep an identifiable record of unfinished work.
Candidate	student with changed requirements	edit task details	correct my plan inputs without replacing task IDs.
Candidate	student working across sessions	report completed minutes or a session percentage	keep the remaining estimate accurate.
Candidate	student revising an estimate	replace remaining work without recording progress	account for newly discovered work.
Candidate	student finishing an assignment	retain a completed task and view completed work	distinguish completion from permanent removal.
Candidate	student correcting mistaken completion	reopen a task with an explicit estimate	return its remaining work to planning.
Candidate	student	permanently delete unwanted tasks	remove work I no longer need to schedule.
Candidate	student with weekly coursework	create a weekly task template	avoid re-entering each occurrence.
Candidate	student with fixed classes or meetings	manage weekly, bounded and one-time commitments	prevent study sessions from overlapping busy time.
Candidate	student whose class is cancelled	skip one commitment occurrence	release that date without deleting the series.
Candidate	unexpectedly busy student	block one date through a one-time commitment	move study work into other available periods.
Candidate	student with a regular routine	configure weekday hours and date overrides	plan within realistic study windows.
Candidate	student needing rest	configure work limits and breaks	avoid excessive continuous study.
Candidate	student balancing deadlines	generate a plan and view daily schedules	see when to work and which work does not fit.
Candidate	returning student	restore records, settings and the exact saved plan	continue without re-entry or a changed schedule.
Candidate	student encountering invalid input or a rejected save	receive an error without partial changes	correct and safely retry the request.
Candidate	student unavailable across multiple dates	submit a continuous or daily unavailable range	block each affected date atomically without entering it separately.

The range story is supported by `CommandDispatcherTest`, `CommitmentServiceTest` and active UI-022 through UI-027. Candidate status describes implemented source; it does not certify completion of storage validation, PDF export or hosted CI.

Appendix C: Non-Functional Requirements

These are quality requirements and operating constraints for the candidate, separate from functional commands and from claims that every platform has been tested. Verification entry points indicate how each requirement can be assessed.

ID	Requirement / operating constraint	Verification entry point
NFR-1	Build and run with Java 25. A packaged JAR requires no Gradle, Node or network connection for normal use.	Java/Gradle version checks, packaged launch and local-session tests.
NFR-2	Read/write saved data and console streams as UTF-8; preserve supported Unicode names through restart. Keep Java sources UTF-8.	Codec, file round-trip and `MainTest` encoding cases.
NFR-3	Save each changed candidate atomically before publishing it or reporting success. Reject a failed save without committing its records, counters or replacement plan.	FileStorage real-file replacement tests, controlled write/move/cleanup faults, and dispatcher/App rollback/retry tests. Injected faults do not certify actual providers or power-loss recovery.
NFR-4	Reject malformed or unreadable saved data before accepting commands; preserve the original file and avoid partial restoration.	Corruption/invalid-UTF-8 tests and Appendix E recovery recipes.
NFR-5	Validate public inputs with runtime checks; use assertions only for internal invariants. Invalid commands must preserve prior state.	Domain/parser/dispatcher tests with invalid and boundary inputs.
NFR-6	Keep default console diagnostics quiet and make stage logging opt-in. Storage stage records must omit saved values and exception payloads.	Logger-capture tests and quiet-stderr console checks; review broader application logs before sharing.
NFR-7	Use one running instance per saved-data file on a filesystem supporting atomic replacement. Close the app before backup/transfer.	Explicit deployment assumption, not a concurrency or universal-provider test result.

No response-time, large-dataset capacity, universal operating-system support or crash/power-loss durability result is claimed. Atomic visibility alone does not flush data to durable media. Current Windows verification cannot establish other platform/provider behavior; see storage verification.

Appendix D: Glossary

- *Task* - A retained item with an optional deadline and exact remaining work; zero remaining minutes means complete.
- *Commitment* - A fixed busy interval that repeats on one weekday, optionally within inclusive dates, or occurs once on a specific date. Planning never moves it.
- *Snapshot* - Immutable plan source data: tasks, commitments, study hours, recurrence templates, optional policy and independent next-ID counters.

- *Plan* - A frozen result derived from one snapshot, containing scheduled sessions, reserved breaks and unallocated work. Successful saved changes invalidate it; policy no-ops preserve it.
- *Feasible allocation* - Sessions and breaks within frozen availability and task timing constraints, with exact work conserved; it need not match the greedy generator's placement or maximize completed tasks.
- *Frozen plan* - The generation time, source snapshot and allocation retained for schedule views until replanning or a successful saved change.
- *Planning horizon* - The calendar range ending at midnight on the same date next year, with February 29 adjusted to February 28 when necessary.
- *Study window* - The continuous daily interval available for study. A date override takes precedence over weekday hours, which take precedence over the default.
- *Cutoff* - The first eligible half-hour boundary on the generation date, rounded up from the current local time and limited to that date's study window.
- *Slot* - One 30-minute interval of reserved scheduling capacity.
- *Exact work* - Minutes credited against a task's estimate; this can be less than the capacity reserved by its final slot.
- *Reserved capacity* - Elapsed time occupied by allocated slots, measured in 30-minute units even when exact task work uses fewer minutes.
- *Recurrence template* - A weekly rule with a stable R ID, first deadline, optional inclusive end date and generation cursor.
- *Occurrence* - A generated task linked to its template ID and original deadline. It has its own T ID and ordinary task lifecycle.
- *Release time* - One week before an occurrence's original deadline; the planner cannot schedule that occurrence earlier, even if its editable deadline changes.
- *Generation cursor* - The latest weekly deadline already processed for a template, including skipped past dates; it prevents backfilling or recreating processed occurrences.

Appendix E: Instructions for Manual Testing

Use the JAR produced at `build/libs/scheduleflow.jar`, Java 25 and a fresh disposable launch folder for each group. Copy `scheduleflow.jar` there and run `java -ea -jar scheduleflow.jar`. IDs below assume empty saved data. Enter only the commands, without the `scheduleflow>` prompt. After `exit`, retain any evidence you need, then discard the disposable folder.

These are representative manual paths, not an exhaustive input matrix. The User Guide command summary gives the full syntax and the UI fixtures supply exact outputs for additional reproducible cases.

Every supported console command has one primary group below. Saving is automatic, so persistence is tested by restarting after ordinary commands rather than through a save command.

Commands	Group
`help`, `exit`, blank input, end of input, launch arguments	Launch, help, exit
`task add`, `task list`, `task list completed`, `task edit`, `task remaining`, `task progress`, `task complete`, `task reopen`, `task delete`	Tasks

Commands	Group
`commitment add` (weekly, bounded and on/one-time), `commitment list`, `commitment edit`, `commitment skip`, `commitment delete`, `unavailable continuous`, `unavailable daily`	Commitments
`plan`, `schedule today`, `schedule DATE`, `hours` set/show/reset, `policy` set/show/reset, `task recur add/list/delete`	Planning
Restart, saved IDs/estimates/settings/templates/plan, missing or invalid data, failed saves	Saving and recovery

Launch, help, exit and console behavior

Use Java 25 and a disposable folder containing the JAR copied from `build/libs/scheduleflow.jar`. Launch from a terminal with `java -ea -jar scheduleflow.jar`, without extra arguments. `-ea` enables assertions; Gradle/distribution launchers already enable them. These paths assume empty saved data and default logging.

Ordered action	Expected result
Launch, then enter `help`.	No banner; exact `scheduleflow>` prompt, then `=== ScheduleFlow Command Help ===` and supported formats, followed by another prompt.
Press Enter on an empty line, then enter a spaces-only line.	Each line produces another prompt with no response text.
Enter `help extra`, then `help`.	First prints `Error: This command accepts no arguments` and `Expected: help`; the corrected command works.
Enter `exit extra`, then `exit`.	The extra argument is rejected with `Expected: exit`; corrected exit prints `Goodbye for now!` once, returns status 0 and produces no further prompt.
Check the disposable folder, then relaunch.	Help, blanks, rejected syntax and exit created no data file; startup is still empty.
Send EOF at the prompt (Ctrl+Z then Enter in a Windows terminal; Ctrl+D on an empty line in a Unix terminal).	Same goodbye and status 0; no save.

In another empty launch folder, run `java -ea -jar scheduleflow.jar --help`. Expect `Error: Usage: java -jar scheduleflow.jar (no arguments)`, status 1 and no prompt. Use `echo $LASTEXITCODE` in PowerShell, or `echo $?` in a Unix shell, immediately after the Java process to inspect its status.

For Unicode and domain-error recovery, start a fresh normal session and run:

```
commitment add n/学习 Café day/TUE start/1000 d/30
commitment add n/Bad day/TUE start/1015 d/30
commitment list
exit
```

Expect C1 with its full Unicode name and 10:00–10:30 interval. The second command reports a half-hour-alignment error and another prompt; the list contains only C1. Relaunch and list again to check the same characters survive persistence. Normal stdout contains no stack traces and stderr is empty. A terminal must also use UTF-8 and a font supporting those characters to display them correctly. Detailed syntax/domain cases are in task testing and commitment testing; fatal startup and rejected saves are covered in storage testing. Exit before discarding the disposable folders; keep your real data untouched.

Output-device failures are checked deterministically through injected writers, not by relying on platform-specific broken pipes. Run the focused `ConsoleUiTest`, `AppTest` and `MainTest` command under console verification. Expect status 1 after a failed prompt without an input read, and after a failed response without another prompt, read or error-output retry. Startup/input/command-error and goodbye writes have the same failure boundary. The saved-mutation test reloads real temporary storage to confirm the task/counter and plan invalidation survive an undeliverable confirmation. These tests do not establish real terminal/provider behavior on another OS.

Tasks and input validation

Use Java 25 and a JAR built from the candidate source described in Task design, not v1.0. Launch `java -ea -jar scheduleflow.jar` from an empty disposable folder. Choose `D` as tomorrow's local date and `E` as the following day, both `YYYY-MM-DD` and within the next year. Replace `D`/`E` before entering commands; they are not special app tokens. Start with no other records, so the first task is T1.

One continuous task lifecycle recipe

```
task add n/Draft due/D 1800 d/90
task list
task edit T1 n/Final draft due/E 1800
task remaining T1 remaining/60
task progress T1 done/15
task progress T1 session/61 percent/25
task progress T1 done/31
task complete T1
task list completed
task progress T1 done/0
task reopen T1 remaining/45
task list completed
task delete T1
task list
task add n/Next due/D 1800 d/1
```

Expect T1 with 90 minutes; the edit retains T1 and 90, changing its name/deadline. The revision leaves 60, direct credit leaves 45, and the session credits 15, leaving 30. `done/31` must reject overreporting without changing 30. Completion prints `Completed T1: Final draft`; the completed list shows T1 at zero. Progress on that completed record must reject even zero credit. Reopening prints `Reopened T1: Final draft | Remaining: 45 min`; the completed view is now empty. Deletion leaves `No tasks found.` The next addition is T2, not reused T1, with a valid one-minute estimate. No timer subtracts work between commands.

Rejection and no-op checks

With active T2, try `task delete T02`, `task edit T2`, `task edit T2 due/2026-02-30 1800`, `task add n/Bad d/30`, `task progress T2 session/60 percent/101`, and `task progress T2 done/2147483648`. Each must show `Error:` and `Expected:` and return to the prompt. `task delete T999` is a value error with no usage line. `task reopen T2 remaining/45` must reject the active task. Check `task list`: T2 still has its original name, deadline and one minute.

Run `plan`, record `schedule today`'s timestamp/rows, then enter `task progress T2 session/1 percent/1` and `task remaining T2 remaining/1`. The first credits zero and the second is unchanged; another schedule view must retain the same plan. A changed `remaining/2` must clear it. Repeating completion of an already-completed task similarly preserves an existing plan. See storage testing for failed-save/restart recipes and Task syntax for more fields. Exit before discarding only this disposable folder. Live timing is not a fixed-clock test.

Retained generated task after template deletion

In a fresh disposable store, add `task recur add n/Quiz due/D 1800 d/30 every/WEEKLY until/D`, then enter `task recur delete R1`. `task recur list` must be empty while `task list` still contains T1. Record `task progress T1 done/7`, complete T1, reopen it with `task reopen T1 remaining/45`, then rename it using `task edit T1 n/Independent quiz`. The same T1 must remain with 45 minutes; the template list stays empty. Restart and check both lists again. The fixed-time `TaskRecurrenceTest` additionally checks original release/provenance, priority, position, settings and counters.

Commitments and unavailable periods

Run each part in a separate empty folder. Parts 1 and 2 use fixed dates because they test calendar membership, not live-clock allocation; commitments do not need future dates. Part 3 uses the live date.

Part 1: lifecycle and conflicts

Ordered command	Expected result
`commitment add n/Lecture day/MON start/1000 d/60 from/2026-11-02 until/2026-11-30`	`Added C1: Lecture | Monday 10:00 - 11:00 (60 min) | 2026-11-02 to 2026-11-30`
`commitment edit C1 start/1030`	`Edited C1: Lecture | Monday 10:30 - 11:30 (60 min) | 2026-11-02 to 2026-11-30`; same ID.
`commitment add n/Clash day/MON start/1100 d/30`	`Error: Commitment overlaps C1: Lecture`; no ID consumed.
`commitment skip C1 on/2026-11-02`	`Skipped C1 on 2026-11-02: Lecture`
`commitment add n/Unavailable on/2026-11-02 start/1030 d/60`	`Added C2: Unavailable | One-time on 2026-11-02 (Monday) 10:30 - 11:30 (60 min)`. It fits because that Monday's lecture is skipped.
`commitment list`	A weekly section with `C1. 10:30 - 11:30 | Lecture | 2026-11-02 to 2026-11-30 | skipped: 2026-11-02`, then a one-time section with C2.
`commitment delete C2`, then `commitment delete C1`	`Deleted commitment C2: Unavailable`, then `Deleted commitment C1: Lecture`.
`commitment list`	`No commitments found.`

Part 2: validation and restart

Ordered command	Expected result
`commitment add n/Tutorial day/WED start/0915 d/60`	`Error: Commitment start must be on a half-hour boundary.`
`commitment add n/Tutorial day/WED start/0900 d/45`	`Error: Duration must be a positive multiple of 30 minutes.`

Ordered command	Expected result
`commitment add n/Tutorial day/WED start/0900 d/60`	Adds C1 (the rejections consumed no ID).
`commitment add n/Lab day/WED start/1000 d/60`	Adds C2; adjacent intervals do not overlap.
`commitment edit C1`	`Error: Provide at least one commitment field to edit.`
`commitment edit C9 start/1200`	`Error: Unknown commitment ID: C9`
`commitment add n/Exam on/2026-11-04 start/1400 d/120`	Adds one-time C3 on Wednesday 2026-11-04.
`commitment skip C3 on/2026-11-04`	`Error: C3 is a one-time commitment; use commitment delete instead.`
`commitment edit C3 day/THU`	`Error: C3 is a one-time commitment; use on/YYYY-MM-DD to change its date.`
`commitment skip C1 on/2026-11-05`	`Error: C1 has no occurrence on 2026-11-05.` (a Thursday)
`commitment add n/Bad day/MON start/1000 d/60 from/2026-11-03 until/2026-11-08`	`Error: Commitment date range must contain its weekday.`
`commitment add n/Bad day/MON start/1000 d/60 from/2026-11-09 until/2026-11-02`	`Error: Commitment start date must not be after its end date.`
`exit`, relaunch in the same folder, then `commitment list`	C1, C2 and C3 are listed unchanged.

Part 3: unavailable period and plan invalidation

Replace `D` with tomorrow's date and `E` with the day after, both as `YYYY-MM-DD`.

Ordered command	Expected result
`task add n/Essay due/E 2200 d/60`, then `plan`	Adds T1 and generates a plan.
`commitment add n/Unavailable on/D start/0800 d/840`	Adds one-time C1 covering the default 08:00–22:00 window.
`schedule D`	`No current plan. Run plan to generate one.` The addition cleared the plan.
`plan`, then `schedule D`	The whole day shows `08:00 - 22:00 | [BUSY] C1: Unavailable`, with no T1 session. T1 is placed in another free slot (today's remaining hours or E, depending on launch time).
`commitment delete C1`, then `schedule D`	No current plan again until `plan` runs.

Part 4: multi-date unavailable batches

Use a fourth empty folder and enter:

```
plan
unavailable continuous n/Trip from/2026-10-26 1500 until/2026-10-28 1100
schedule today
exit
```

Expect one confirmation containing C1 for 26 October 15:00–24:00, C2 for the full 27 October and C3 for 28 October 00:00–11:00. The schedule command must say there is no current plan because the successful batch invalidated it. Relaunch and use `commitment list`; all three one-time entries must remain.

In another empty folder, run `unavailable daily n/Appointment from/2026-10-26 until/2026-10-28 start/1500 end/1700`. Expect three dated records with the same 15:00–17:00 window and consecutive IDs. Delete C2 and confirm C1 and C3 remain independently available.

For atomic rejection, first add a one-time commitment on 27 October from 16:00–16:30, then attempt that daily batch. Expect an overlap error, no generated periods and an unchanged next ID; a following valid addition must receive C2. UI-024 supplies the exact reversed/equal range, overnight daily, alignment and syntax rejections. Deterministic dispatcher tests cover a failed save and insufficient ID capacity, which are not practical to force through an ordinary JAR session.

Planning, schedules, hours, policy and recurrence

Direct malformed-plan validation is an API check covered by `PlannerTest`; the console has no command for supplying raw allocations. Run those fixed-time tests to check rejection independently of live-clock manual scenarios.

Use another empty folder. Choose `D` as tomorrow and `W` as its full lowercase weekday (for example `tuesday`); substitute them in the commands.

Ordered command	Expected result
`schedule today`	No current plan; asks you to run `plan`.
`hours W /start 09:00 /end 17:00`, then `hours W`	Sets and shows recurring weekday hours.
`hours D /start 10:00 /end 12:00`, then `hours D`	Date override wins: 10:00–12:00.
`hours D /start 10:15 /end 12:00`	Alignment error; previous hours retained.
`policy set max/60 break/30`, then `policy show`	Shows the stored work/break limits.
`task add n/Study due/D 1200 d/90`, then `plan`	Generates a full or partial result without reducing the stored 90 minutes.
`schedule D`, then `schedule today`	Shows dates from the frozen plan, its generation time and applicable labels. Exact rows depend on launch time.
`schedule 0001-01-01`	Out-of-range error; current plan remains available.
`policy set max/60 break/30`, then `schedule D`	Reports unchanged policy; preserves the plan.
`policy reset`, then `schedule D`	Restores unrestricted planning; no current plan.
`hours reset D`, then `hours D`	Falls back to weekday 09:00–17:00.
`hours reset W`, then `hours W`	Restores default 08:00–22:00.
`task recur add n/Quiz due/D 1800 d/30 every/WEEKLY until/D`	Adds R1 and its one future T occurrence.

Ordered command	Expected result
`task recur list`, then `task list`	Template has an R ID; occurrence has a separate T ID.
`task recur delete R1`, then `task list`	Stops generation; existing occurrence remains.

A repeatable partial result can be checked in a fresh folder by adding `task add n/Large due/D 0000 d/2147483647`, then running `plan`. Expect unallocated work and unchanged stored minutes. For an exact complete/BREAK schedule, use the controlled-clock User Guide example as a reference, or replay the saved-plan fixtures UI-029/UI-031; the production JAR has no clock-setting flag. Live tests cannot promise those same timestamps.

For a small exact partial-allocation check, use a second empty launch folder. Choose D0 as today and D1 as tomorrow, both within the next year. Replace `D0` and `D1` in the commands with their actual `YYYY-MM-DD` dates. Run `plan` after 08:30 local time on D0 so its short window has passed:

```
hours D0 /start 08:00 /end 08:30
hours D1 /start 08:00 /end 09:00
commitment add n/Busy on/D1 start/0800 d/30
task add n/Partial due/D1 0900 d/120
plan
schedule D0
schedule D1
```

The plan should schedule 30 minutes and report the remaining 90 as unallocated for insufficient capacity; the saved task estimate remains 120 minutes. `schedule D0` shows its past window as unplanned. On D1, the commitment occupies 08:00-08:30 and the only remaining slot is 08:30-09:00. Keep the data folder separate from other manual tests so its IDs and settings are isolated.

Saving, restoration, failure and recovery

Use Java 25 and the candidate `scheduleflow.jar` built from the source described in persistence design. Launch with `java -ea -jar scheduleflow.jar` in disposable folders. Keep one instance per folder. Never use your normal data for these checks; see the UG persistence chapter for file location and recovery. Replace each `D` below with tomorrow's ISO date before entering commands; its 18:00 deadline must be within the allowed calendar-year range.

First launch, restart and frozen plan

1. In folder A, confirm `data/scheduleflow.txt` is absent. Launch and enter `task list`. Expect `No tasks found.` and another prompt; reading alone must not create the data folder.
2. Enter this setup, replacing `D` first:

```
task add n/Draft due/D 1800 d/45
task add n/Remove due/D 1800 d/30
task delete T2
commitment add n/Class day/TUE start/1000 d/30
commitment add n/Remove day/WED start/1200 d/30
commitment delete C2
hours monday /start 09:00 /end 18:00
policy set max/60 break/30
task recur add n/Weekly due/D 1800 d/60 every/WEEKLY until/D
plan
task list
task recur list
schedule today
exit
```

3. Record the selected schedule date, generation timestamp and rows. Expect T1 with 45 minutes, generated T3 with 60 minutes, C1 and template R1. The data file now exists. Copy it to a backup with the app closed.
4. Relaunch A. Check `task list`, `commitment list`, `task recur list`, `hours monday`, `policy show`, and `schedule D0`, replacing `D0` with the recorded schedule date (not tomorrow's `D`). Expect identical records, template cursor, 09:00–18:00 Monday hours, 60/30 policy and the same schedule timestamp/rows. These reads must leave the saved bytes unchanged.
5. Add another ordinary task and commitment using valid inputs. Expect T4 and C3, not reused deleted IDs. Add a second weekly template: expect R2. These successful changes invalidate the plan; `schedule D0` now reports `No current plan. Run plan to generate one.` Exit.

Backup, invalid data and recovery

1. In folder B, place a copy of A's step-3 backup at `data/scheduleflow.txt`. Launch and repeat the step-4 reads. Expect the backed-up values and frozen schedule. Exit before changing the copy.
2. Change only B's header to `SCHEDULEFLOW|99`. Launch. Expect `Error: Unsupported file version: 99`, status 1 and no prompt. Check that the invalid file was not rewritten. Restore the backup and expect normal startup.
3. Repeat with an extra `BROKEN` line appended to the restored plan file. Expect a saved-plan validation error before the prompt and unchanged invalid bytes. Restore the backup again. Exact wording depends on which plan row/count fails. Also try the complete impossible-work fixture in the storage handoff. Its real task ID and valid interval must still fail with `Error: Invalid saved plan: Scheduled work plus remainder must equal source remaining work.`, status 1, no prompt and unchanged bytes. Restore the backup afterward.
4. For an access/type rejection, close the app, move B's data file aside and create a directory named `scheduleflow.txt` in its place. Launch. Expect `Error: Cannot load saved data: ...`, status 1 and no prompt. Remove only that empty test directory and restore the moved file. Permission-denial recipes vary by OS/account; record them as unexecuted if they cannot be reproduced.

Rejected save and retry in one session

1. Launch in fresh folder C and wait for the first prompt. Only then, in a second terminal or file manager, create a regular text file named `data` in C. For PowerShell opened in C, use `Set-Content -LiteralPath data -Value blocked`.
2. In the running app enter `commitment add n/Retry day/TUE start/1000 d/30`. Expect `Error: Cannot save data: ...`, another prompt and no success message. `commitment list` must report `No commitments found.` The blocking file must retain its contents.
3. Move that test file aside while the app remains open, then repeat the same add. Expect C1. Exit, relaunch and confirm C1 is restored. Creating the blocker before startup tests loading instead, so the timing matters.

Close every instance before cleanup. Remove only disposable A/B/C folders and their copies/logs after inspection. This recipe tests a blocked parent, not an unsupported atomic move or failed temporary-file cleanup; those provider failure paths have deterministic injected JUnit coverage, separate from this manual recipe and from actual provider evidence. If cleanup itself fails, an abandoned temporary sibling can remain; preserve the main file and inspect the failure before discarding disposable test artifacts. Codec, filesystem-load and packaged startup regressions verify impossible-plan rejection through the integrated planning validator without changing the original saved file.